



nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation

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Biomedical imaging is a driver of scientific discovery and a core component of medical care and is being stimulated by the field of deep learning. While semantic segmentation algorithms enable image analysis and quantification in many applications, the design of respective specialized solutions is non-trivial and highly dependent on dataset properties and hardware conditions. We developed nnU-Net, a deep learning-based segmentation method that automatically configures itself, including preprocessing, network architecture, training and post-processing for any new task. The key design choices in this process are modeled as a set of fixed parameters, interdependent rules and empirical decisions. Without manual intervention, nnU-Net surpasses most existing approaches, including highly specialized solutions on 23 public datasets used in international biomedical segmentation competitions. We make nnU-Net publicly available as an out-of-the-box tool, rendering state-of-the-art segmentation accessible to a broad audience by requiring neither expert knowledge nor computing resources beyond standard network training.

Semantic segmentation transforms raw biomedical image data into meaningful, spatially structured information and thus plays an essential role in scientific discovery^{1,2}. At the same time, semantic segmentation is an essential ingredient in numerous clinical applications^{3,4}, including applications of artificial intelligence in diagnostic support systems^{5,6}, therapy planning support⁷, intra-operative assistance² and tumor growth monitoring⁸. High interest in automatic segmentation methods manifests in a thriving research landscape, accounting for 70% of international image analysis competitions in the biomedical sector⁹.

Despite the recent success of deep learning-based segmentation methods, their applicability to specific image analysis problems of end-users is often limited. The task-specific design and configuration of a method requires high levels of expertise and experience, with small errors leading to large drops in performance¹⁰. Especially in three-dimensional (3D) biomedical imaging, for which dataset properties such as imaging modality, image size, (anisotropic) voxel spacing and class ratio vary drastically, this process can be cumbersome and successful configurations from one dataset rarely translate to another. The numerous expert decisions involved in adapting and training a neural network range from exact network architecture to training schedule and methods for data augmentation or post-processing. Each of the interdependent sub-components is controlled by essential parameters such as learning rate, batch size or class sampling strategy¹⁰. An additional layer of complexity in the overall setup is posed by the hardware available for training and inference¹¹. Purely empirical optimization of the co-dependent design choices in this high-dimensional space, as proposed by previous studies in the field of automatic machine learning (AutoML)¹², amplifies both the number of required training cases, as well as compute resources, by orders of magnitude and typically covers a mere fraction of the segmentation pipeline (such as the architecture or the data augmentation), leaving a considerable proportion of its configuration to the experimenter¹². Moreover, the application of AutoML to new datasets comes with its own set of required

expert choices, for example, when considering the construction of a sensible problem-specific search space¹². As our analysis of the current landscape in international biomedical segmentation challenges indicates (Results), these practical limitations commonly leave users with a manual and an iterative trial-and-error process during method design that is mostly driven by individual experience, is only scarcely documented and often results in suboptimal segmentation pipelines^{10,13}.

In this work, we outline a new path between the status quo of primarily expert-driven method configuration in biomedical segmentation on one side and primarily data-driven AutoML approaches on the other. Specifically, we define a recipe hereafter that systematizes the configuration process on a task-agnostic level and drastically reduces the search space for empirical design choices when given a new task.

1. Collect design decisions that do not require adaptation between datasets and identify a robust common configuration (‘fixed parameters’).
2. For as many of the remaining decisions as possible, formulate explicit dependencies between specific dataset properties (‘dataset fingerprint’) and design choices (‘pipeline fingerprint’) in the form of heuristic rules to allow for almost-instant adaptation on application (‘rule-based parameters’).
3. Learn only the remaining decisions empirically from the data (‘empirical parameters’).

Our implementation of this recipe was developed and validated on the ten datasets provided by the Medical Segmentation Decathlon. The resulting segmentation method, which we call nnU-Net, is able to perform automated configuration for arbitrary new datasets. In contrast to existing research methods, nnU-Net is holistic, that is, its automated configuration covers the entire segmentation pipeline (including essential topological parameters of the network architecture) without any manual decisions. Moreover, automated configuration in nnU-Net is fast, comprising a simple execution of rules and only a few empirical choices to be made, thus

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nnU-Net: 一种用于基于深度学习的生物医学图像分割的自配置方法

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生物医学成像是科学发现的驱动力，也是医疗保健的核心组成部分，并正受到深度学习领域的推动。尽管语义分割算法能够在许多应用中实现图像分析与量化，但各自专用解决方案的设计并非易事，且高度依赖于数据集属性与硬件条件。我们开发了nnU-Net，一种基于深度学习的分割方法，能够针对任何新任务自动配置自身，包括预处理、网络架构、训练和后处理。该过程中的关键设计选择被建模为一组固定参数、相互依赖的规则和经验决策。无需人工干预，nnU-Net在23个用于国际生物医学分割竞赛的公共数据集上超越了大多数现有方法，包括高度专业化的解决方案。我们将nnU-Net作为开箱即用的工具公开提供，使最先进的分割技术能够为广泛受众所用，既不需要专家知识，也不需要超出标准网络训练的计算资源。

Semantic segmentation将原始生物医学图像数据转化为有意义的、空间结构化的信息，因而在科学发现中发挥着至关重要的作用^{1,2}。与此同时，语义分割也是众多临床应用的重要组成部分^{3,4}，包括人工智能在诊断支持系统^{5,6}、治疗计划支持⁷、术中辅助²以及肿瘤生长监测⁸中的应用。对自动分割方法的高度关注体现在蓬勃发展的研究格局中，在生物医学领域占国际图像分析竞赛的70%⁹。

尽管基于深度学习的分割方法近来取得了成功，但其对最终用户特定图像分析问题的适用性往往有限。方法的任务特定设计与配置需要高水平的专业知识和经验，微小的错误便会导致性能大幅下降¹⁰。尤其是在三维（3D）生物医学成像中，成像模态、图像尺寸、（各向异性）体素间距和类别比例等数据集属性差异极大，这一过程可能极为繁琐，且从一个数据集获得的成功配置鲜少能迁移至另一个数据集。在调整和训练神经网络过程中涉及的大量专家决策，从确切的网络架构到训练计划以及数据增强或后处理的方法，范围广泛。每个相互依赖的子组件都由学习率、批量大小或类别采样策略等关键参数所控制¹⁰。整体设置中额外的复杂性来自于可用于训练和推理的硬件¹¹。正如自动机器学习（AutoML）领域的先前研究所提出的，在这一高维空间中对相互依赖的设计选择进行纯经验性优化¹²，会将所需的训练样本数量以及计算资源放大数个数量级，并且通常仅覆盖分割流水线的一小部分（例如架构或数据增强），将相当大比例的配置留给了实验者¹²。此外，将AutoML应用于新数据集还伴随着其自身所需的一系列

专家选择，例如在考虑构建合理的针对特定问题的搜索空间时¹²。正如我们对当前国际生物医学分割挑战赛现状的分析所表明的（结果），这些实际限制通常使用户在方法设计过程中陷入手动且迭代的试错过程，该过程主要由个人经验驱动，仅有少量文档记录，并且往往导致次优的分割流水线^{10,13}。

在这项工作中，我们在生物医学分割领域主要由专家驱动的方法配置现状与另一方面主要由数据驱动的自动机器学习方法之间，勾勒出一条新的路径。具体而言，我们在此定义了一套方法，将配置过程在任务无关的层面上系统化，并在给定新任务时大幅缩减经验设计选择的搜索空间。

1. 收集不需要在数据集之间进行适配的设计决策，并确定一个稳健的通用配置（“固定参数”）。2. 对于尽可能多的剩余决策，以启发式规则的形式，明确表述特定数据集属性（“数据集指纹”）与设计选择（“流水线指纹”）之间的依赖关系，以便在应用时几乎即时完成适配（“基于规则的参数”）。3. 仅从数据中经验性地学习剩余的决策（“经验参数”）。

我们对该配方的实现是在医学分割十项全能提供的十个数据集上开发和验证的。由此产生的分割方法，我们称之为nnU-Net，能够对任意新数据集执行自动配置。与现有的研究方法相比，nnU-Net是整体性的，即其自动配置覆盖了整个分割流水线（包括网络架构的基本拓扑参数），无需任何手动决策。此外，nnU-Net中的自动配置速度很快，仅需简单执行规则和少量经验性选择即可，因此

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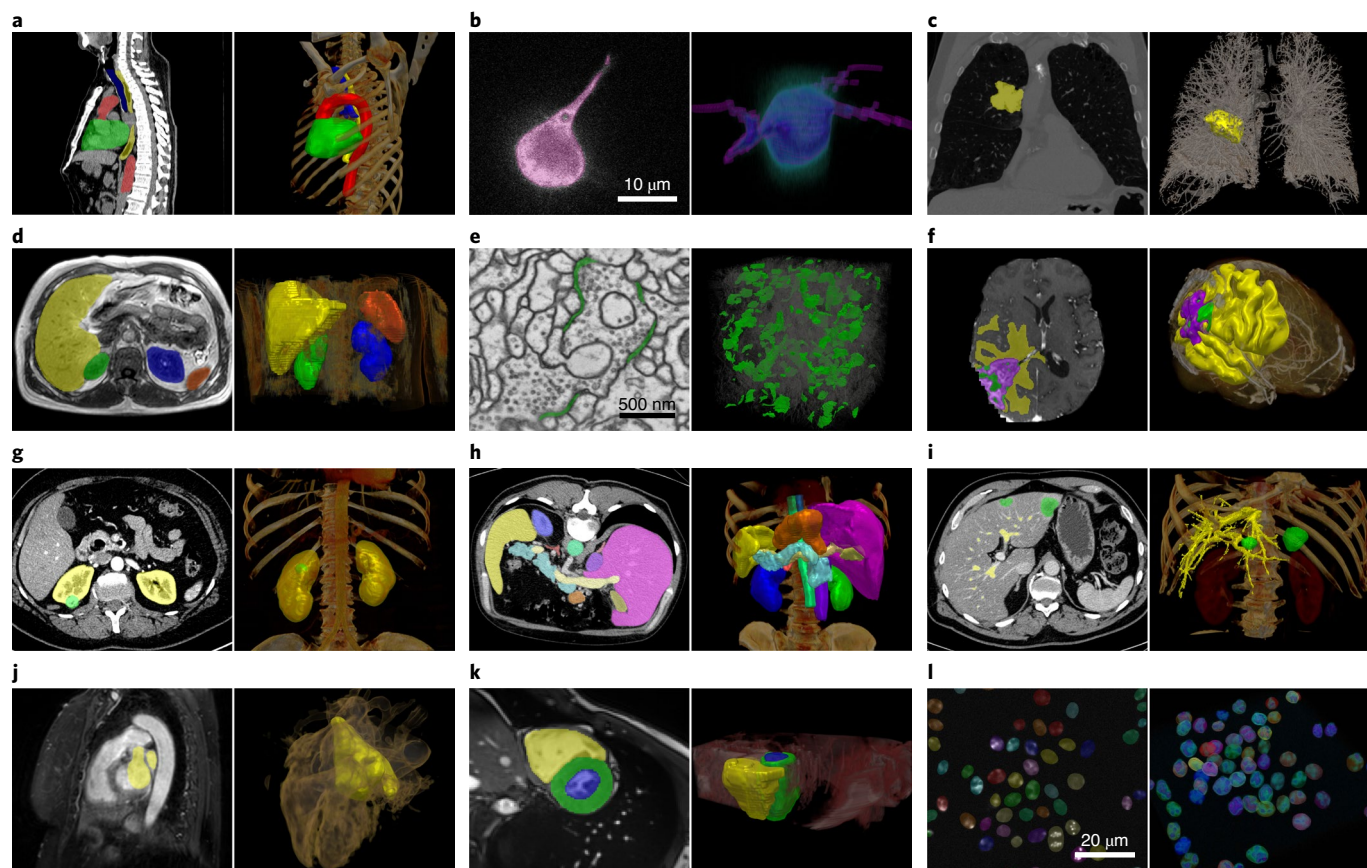


Fig. 1 | nnU-Net handles a broad variety of datasets and target image properties. All examples originate from test sets of different international segmentation challenges to which nnU-Net was applied. Target structures for each dataset are shown in 2D projected onto the raw data (left) and in 3D together with a volume rendering of the raw data (right). All visualizations were created with the MITK Workbench³⁵. **a**, Heart (green), aorta (red), trachea (blue) and esophagus (yellow) in CT images (dataset (D)18)¹⁴. **b**, A549 lung cancer cells (purple) in FM (D22)^{36,37}. **c**, Lung nodules (yellow) in CT images (D6)¹⁴. **d**, Liver (yellow), spleen (orange), left and right kidneys (blue and green, respectively) in T1 in-phase MRI (D16)²⁰. **e**, Synaptic clefts (green) in EM scans (D19) (<https://cremi.org/>). **f**, Edema (yellow), enhancing tumor (purple), necrosis (green) in MRI (T1, T1 with contrast agent, T2, FLAIR) (D1)^{14,38}. **g**, Kidneys (yellow) and kidney tumors (green) in CT images (D17)²¹. **h**, Thirteen abdominal organs in CT images (D11)¹⁶. **i**, Hepatic vessels (yellow) and liver tumors (green) in CT (D8)¹⁴. **j**, Left ventricle (yellow) in MRI (D2)¹⁴. **k**, Right ventricle (yellow), left ventricular cavity (blue) and myocardium of left ventricle (green) in cine MRI (D13)⁶. **l**, HL60 cell nuclei (instance segmentation, one color per instance) in FM (D21)³⁹.

requiring virtually no compute resources beyond standard model training. Lastly, nnU-Net is data efficient; encoding design choices based on a large and diverse data pool serves as a strong inductive bias for application to datasets with limited training data.

The general applicability of nnU-Net's automated configuration is demonstrated in 13 additional datasets. Altogether, we report results on 53 segmentation tasks, covering an unprecedented diversity of target structures, image types and image properties. As an open source tool, nnU-Net can simply be trained out-of-the box to generate state-of-the-art segmentations.

Results

nnU-Net is a deep learning-based segmentation method that automatically configures itself, including preprocessing, network architecture, training and post-processing for any new task in the biomedical domain. Exemplary segmentation results generated by nnU-Net for a variety of datasets are shown in Fig. 1.

nnU-Net automatically adapts to any new dataset. Fig. 2 shows how nnU-Net systematically addresses the configuration of entire segmentation pipelines and provides a visualization and description of the most relevant design choices.

nnU-Net development. The automatic configuration of nnU-Net is based on distilling domain knowledge into three parameter groups: fixed, rule-based and empirical parameters. First, we collect all design choices that do not require adaptation between datasets (such as setting the architecture template to 'U-Net-like') and optimize their joint configuration for robust generalization on our development datasets. Second, for as many of the remaining decisions as possible, we formulate explicit dependencies between the 'dataset fingerprint', a standardized dataset representation comprising key properties such as image size, voxel spacing information or class ratios, and the 'pipeline fingerprint', which we define as the entirety of choices being made during method design. The dependencies are modeled in the form of interdependent heuristic rules that allow for almost-instant execution upon application. To give an example for such rules, the interdependent configuration of batch size, patch size and network topology is based on the following three principles.

- Larger batch sizes allow for more accurate gradient estimates and are thus preferable (up to a sweet spot typically not reached in our domain), but in practice any batch size larger than one already results in robust training.

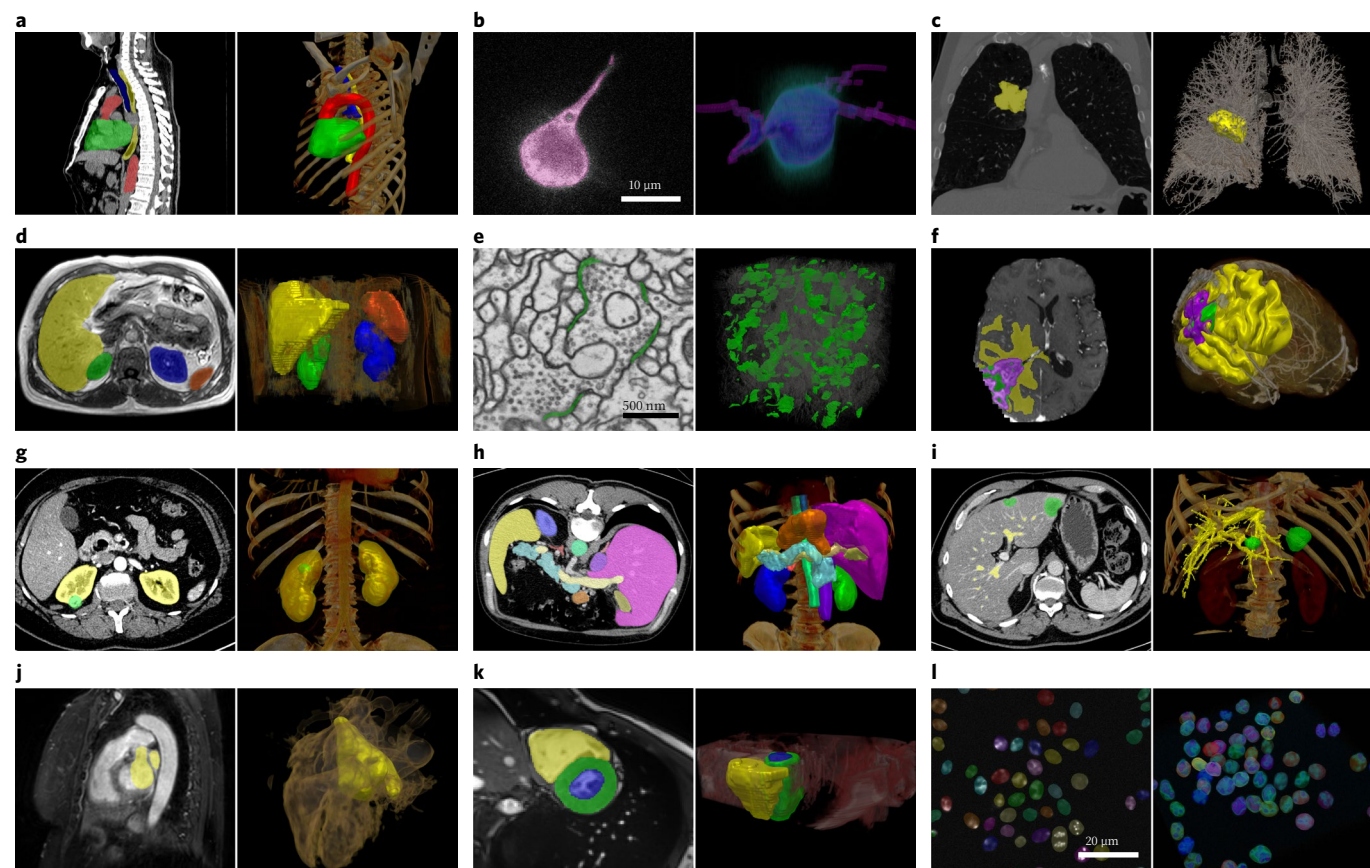


图 1 | nnU-Net可处理多种数据集和目标图像属性。所有示例均来自nnU-Net所应用的不同国际分割挑战赛的测试集。每个数据集的目标结构以二维投影方式叠加在原始数据上展示（左），并以三维方式与原始数据的体绘制一同展示（右）。所有可视化均使用MITK Workbench创建³⁵。a, CT图像（数据集(D)18）中的心脏（绿色）、主动脉（红色）、气管（蓝色）和食管（黄色）¹⁴。b, 荧光显微镜（D22）中的A549肺癌细胞（紫色）^{36,37}。c, CT图像（D6）中的肺结节（黄色）¹⁴。d, T1同相位MRI（D16）中的肝脏（黄色）、脾脏（橙色）、左右肾脏（分别为蓝色和绿色）²⁰。e, 电子显微镜扫描（D19）中的突触间隙（绿色）（<https://cremi.org/>）。f, MRI（T1、增强T1、T2、FLAIR）（D1）中的水肿（黄色）、增强肿瘤（紫色）、坏死（绿色）^{14,38}。g, CT图像（D17）中的肾脏（黄色）和肾脏肿瘤（绿色）²¹。h, CT图像（D11）中的十三种腹部器官¹⁶。i, CT（D8）中的肝血管（黄色）和肝肿瘤（绿色）¹⁴。j, MRI（D2）中的左心室（黄色）¹⁴。k, 电影MRI（D13）中的右心室（黄色）、左心室腔（蓝色）和左心室心肌（绿色）⁶。l, 荧光显微镜（D21）中的HL60细胞核（实例分割，每个实例一种颜色）³⁹。

几乎不需要超出标准模型训练的计算资源。最后，nnU-Net具有数据高效性；基于庞大且多样化的数据池来编码设计选择，为应用于训练数据有限的数据集提供了强大的归纳偏置。

nnU-Net自动配置的普遍适用性在另外13个数据集上得到了验证。总体而言，我们报告了53个分割任务的结果，涵盖了前所未有的目标结构、图像类型和图像属性的多样性。作为一个开源工具，nnU-Net可以直接开箱即用地进行训练，以生成最先进的分割结果。

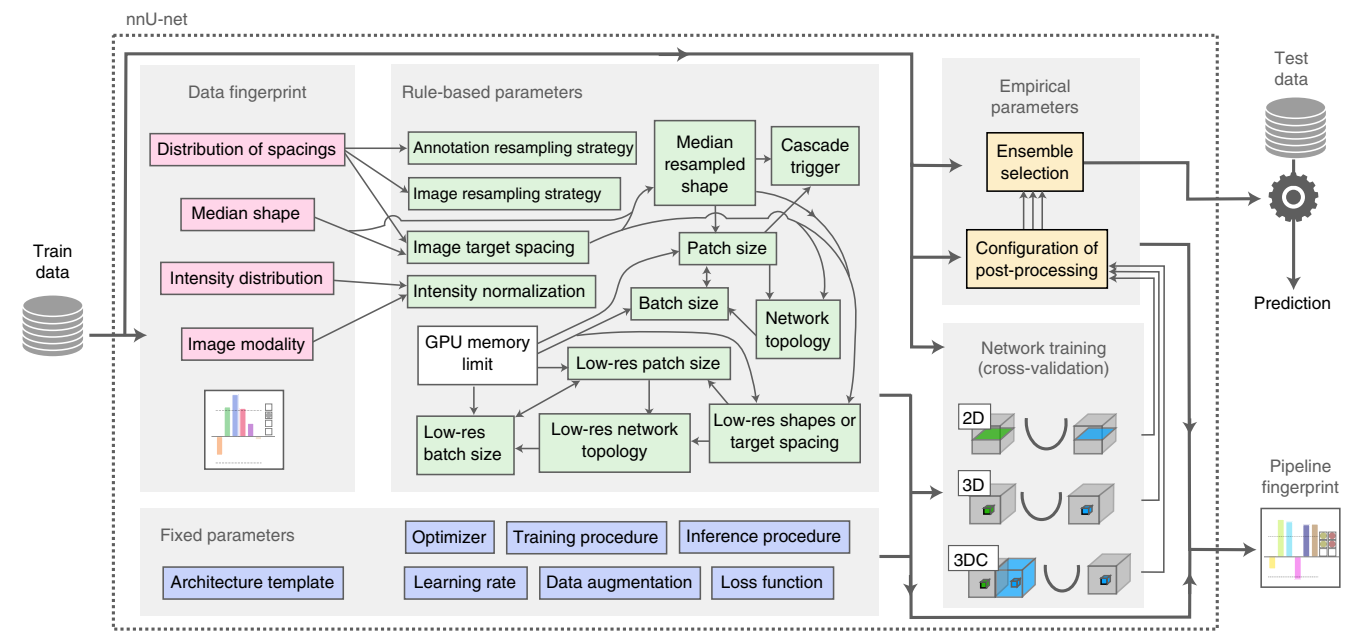
结果

nnU-Net是一种基于深度学习的分割方法，能够自动配置自身，包括针对生物医学领域中任何新任务的预处理、网络架构、训练和后处理。图1展示了nnU-Net为各种数据集生成的示例性分割结果。

nnU-Net能够自动适应任何新数据集。图2展示了nnU-Net如何系统性地处理整个分割流水线的配置，并提供了最相关设计选择的可视化与描述。

nnU-Net的开发。nnU-Net的自动配置基于将领域知识蒸馏为三个参数组：固定参数、基于规则的参数和经验参数。首先，我们收集所有不需要在数据集之间进行适配的设计选择（例如将架构模板设置为“类U-Net”），并针对我们在开发数据集上的稳健泛化来优化其联合配置。其次，对于剩余决策中尽可能多的部分，我们明确制定了“数据集指纹”（一种标准化的数据集表示，包含图像尺寸、体素间距信息或类别比例等关键属性）与“流水线指纹”（我们将其定义为方法设计过程中所做出的全部选择的集合）之间的显式依赖关系。这些依赖关系以相互依赖的启发式规则的形式进行建模，使得在应用时几乎可以即时执行。举一个此类规则的例子，批量大小、图像块大小和网络拓扑的相互依赖配置基于以下三条原则。

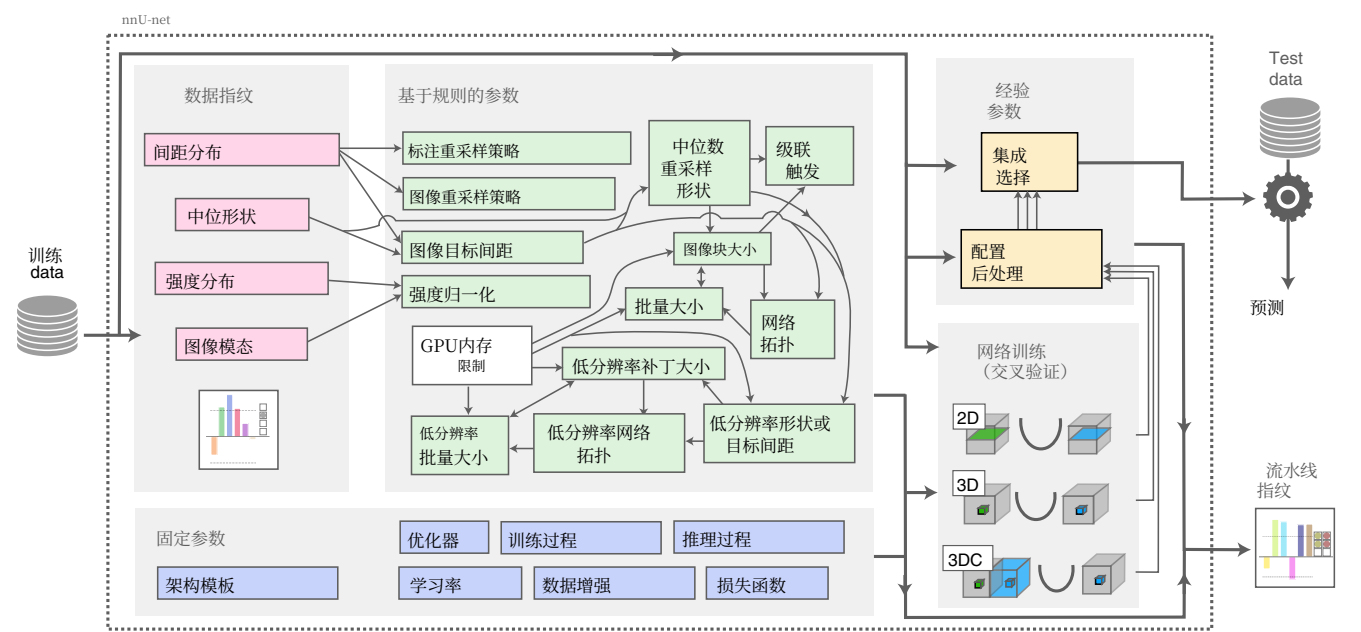
- 更大的批量大小允许更准确的梯度估计，因此更可取（直到达到一个在我们的领域中通常达不到的最佳点），但在实践中，任何大于一的批量大小已经能够实现稳健的训练。



Design choice	Required input	Automated (fixed, rule-based or empirical) configuration derived by distilling expert knowledge (more details in online methods)	Image target spacing	Distribution of spacings	If anisotropic, lowest resolution axis tenth percentile, other axes median. Otherwise, median spacing for each axis. (computed based on spacings found in training cases)
Learning rate	–	Poly learning rate schedule (initial, 0.01)	Network topology, patch size, batch size	Median resampled shape, target spacing, GPU memory limit	Initialize the patch size to median image shape and iteratively reduce it while adapting the network topology accordingly until the network can be trained with a batch size of at least 2 given GPU memory constraints. for details see online methods.
Loss function	–	Dice and cross-entropy	Trigger of 3D U-Net cascade	Median resampled image size, patch size	Yes, if patch size of the 3D full resolution U-Net covers less than 12.5% of the median resampled image shape
Architecture template	–	Encoder–decoder with skip-connection ('U-Net-like') and instance normalization, leaky ReLU, deep supervision (topology-adapted in inferred parameters)	Configuration of low-resolution 3D U-Net	Low-res target spacing or image shapes, GPU memory limit	Iteratively increase target spacing while reconfiguring patch size, network topology and batch size (as described above) until the configured patch size covers 25% of the median image shape. For details, see online methods.
Optimizer	–	SGD with Nesterov momentum ($\mu = 0.99$)	Configuration of post-processing	Full set of training data and annotations	Treating all foreground classes as one; does all-but-largest-component-suppression increase cross-validation performance? Yes, apply; reiterate for individual classes No, do not apply; reiterate for individual foreground classes
Data augmentation	–	Rotations, scaling, Gaussian noise, Gaussian blur, brightness, contrast, simulation of low resolution, gamma correction and mirroring	Ensemble selection	Full set of training data and annotations	From 2D U-Net, 3D U-Net or 3D cascade, choose the best model (or combination of two) according to cross-validation performance
Training procedure	–	1,000 epochs $\times$ 250 minibatches, foreground oversampling			
Inference procedure	–	Sliding window with half-patch size overlap, Gaussian patch center weighting			
Intensity normalization	Modality, intensity distribution	If CT, global dataset percentile clipping & z score with global foreground mean and s.d. Otherwise, z score with per image mean and s.d.			
Image resampling strategy	Distribution of spacings	If anisotropic, in-plane with third-order spline, out-of-plane with nearest neighbor Otherwise, third-order spline			
Annotation resampling strategy	Distribution of spacings	Convert to one-hot encoding $\rightarrow$ If anisotropic, in-plane with linear interpolation, out-of-plane with nearest neighbor Otherwise, linear interpolation			

Fig. 2 | Proposed automated method configuration for deep learning-based biomedical image segmentation. Given a new segmentation task, dataset properties are extracted in the form of a ‘dataset fingerprint’ (pink). A set of heuristic rules models parameter interdependencies (shown as thin arrows) and operates on this fingerprint to infer the data-dependent ‘rule-based parameters’ (green) of the pipeline. These are complemented by ‘fixed parameters’ (blue), which are predefined and do not require adaptation. Up to three configurations are trained in a five-fold cross-validation. Finally, nnU-Net automatically performs empirical selection of the optimal ensemble of these models and determines whether post-processing is required (‘empirical parameters’, yellow). The table on the bottom shows explicit values as well as summarized rule formulations of all configured parameters. Res., resolution.

- Larger patch size during training increases the contextual information absorbed by the network and is thus crucial for performance.
- The topology of a network should be deep enough to guarantee an effective receptive field size at least as large as the patch size, so that contextual information is not discarded.



设计选择	必需输入	自动化 (固定、基于规则或经验) 通过提炼专家知识得出的配置知识 (更多细节见在线方法)	图像目标间距	分布间距	若为各向异性, 取最低分辨率轴的第十百分位数, 其他轴取中位数。 否则, 每个轴取间距的中位数。 (根据训练样本中的间距计算)
学习率	–	Poly学习率调度 (初始值0.01)	网络拓扑, 图像块大小, 批量大小	中位数重采样形状、目标间距、GPU内存限制	将图像块大小初始化为中位数图像形状, 并在调整网络的同时迭代减小它相应地调整拓扑, 直到网络可以被训练在给定GPU内存的情况下, 批量大小至少为2限制。详情见在线方法。
损失函数	–	Dice与交叉熵	3D的触发 U-Net 级联	中位数重采样图像尺寸, 图像块大小	是, 如果3D全分辨率U-Net的图像块大小覆盖不到中位数重采样的12.5%图像形状
架构模板	–	带跳跃连接的编码器-解码器 (“类U-Net”) 以及实例归一化、Leaky ReLU、深度监督 (在推断参数中适配拓扑)	配置低分辨率 3D U-Net	低分辨率目标间距或图像形状、GPU内存限制	在重新配置的同时迭代增加目标间距图像块大小、网络拓扑和批量大小 (如上文所述), 直到配置的图像块大小覆盖了中位数图像形状的25%。有关详细信息, 请参阅在线方法。
优化器	–	带Nesterov动量的SGD ($\mu = 0.99$)	配置后处理的	全套训练数据 and 标注	将所有前景类别视为一类; 是否除最大组件抑制外-最大组件抑制能提升交叉-验证性能? 是的, 应用; 针对各个类别重复此操作不, 不应用; 对各个前景类别重复
Data数据增强	–	旋转、缩放、高斯噪声、高斯模糊、亮度、对比度、低分辨率模拟、伽马校正和镜像	标注重采样策略	分布 of 间距	转换为独热编码 $\rightarrow$ 如果为各向异性, 平面内使用线性插值, 平面外使用最近邻 否则, 使用线性插值
训练过程	–	1,000 轮次 $\times$ 250 小批量、前景过采样			
推理过程	–	采用半图块尺寸重叠的滑动窗口, 高斯图块中心加权			
强度归一化	模态, 强度分布	如果是计算机断层扫描, 则使用全局数据集百分位裁剪和 z 基于全局前景均值和标准差的评分 否则, 使用每图像均值和标准差的 z 评分			
图像重采样策略	分布 of 间距	若为各向异性, 平面内采用三阶样条, 平面外-平面外使用最近邻 否则, 使用三阶样条			
标注重采样策略	分布 of 间距	转换为独热编码 $\rightarrow$ 如果为各向异性, 平面内使用线性插值, 平面外使用最近邻 否则, 使用线性插值			
集成选择	–	从 2D U-Net、3D U-Net 或 3D 级联中, 选择根据交叉验证性能选择最佳模型 (或两者的组合) 验证性能			

图 2 | 提出的基于深度学习的生物医学图像分割自动化方法配置。 给定一个新的分割任务, 以“数据集指纹”的形式提取数据集属性 (粉色)。一组启发式规则对参数相互依赖关系进行建模 (以细箭头表示), 并基于该指纹进行操作, 以推断流水线中依赖于数据的“基于规则的参数” (绿色)。这些参数由“固定参数” (蓝色) 补充, 固定参数是预定义的, 无需自适应调整。最多有三种配置在五折交叉验证中进行训练。最后, nnU-Net自动对这些模型的最优集成进行经验选择, 并确定是否需要后处理 (“经验参数”, 黄色)。底部的表格显示了所有已配置参数的显式值以及归纳出的规则表述。Res., 分辨率。

- 训练期间更大的图像块大小会增加网络吸收的上下文信息, 因此对性能至关重要。
- 网络的拓扑结构应足够深, 以确保有效感受野大小至少与图像块大小相当, 从而使上下文信息不被丢弃。

Distilling this knowledge into successful method design results in the following heuristic rule: “initialize the patch size to median image shape and iteratively reduce it while adapting the network topology accordingly (including network depth, number and position of pooling operations along each axis, feature map sizes and convolutional kernel sizes) until the network can be trained with a batch size of at least two given GPU memory constraints.” A detailed description of all heuristic rules is provided in the online methods, and a compilation of guiding principles used for the derivation of rules is provided in Supplementary Note 2. Third, we set only the remaining design choices, that is, model selection and post-processing, to be decided empirically based on the training data during application. Our implementation of this recipe, which we call nnU-Net, was exclusively developed on a set of ten development datasets originating from the Medical Decathlon Segmentation Challenge¹⁴.

nnU-Net application. When applying nnU-Net to a new dataset, the automatic configuration of nnU-Net runs without manual intervention. Thus, with the exception of the few remaining empirical choices to be made, no additional computational cost beyond a standard network training procedure is required. nnU-Net’s automated method configuration starts with the extraction of the dataset fingerprint and subsequent execution of heuristic rules. By default, nnU-Net generates three different U-Net¹⁵ configurations: a two-dimensional (2D) U-Net, a 3D U-Net that operates at full image resolution and a 3D U-Net cascade in which the first U-Net operates on downsampled images, and the second is trained to refine the segmentation maps created by the former at full resolution. After cross-validation, nnU-Net empirically chooses the best performing configuration or ensemble. Finally, nnU-Net empirically opts for ‘non-largest component suppression’ as a post-processing step if performance gains are measured. The output of nnU-Net’s automated configuration and training process are fully trained models that can be deployed to make predictions on unseen images. We demonstrate the generalization ability of the design choices encoded in nnU-Net’s fixed, rule-based and empirical parameters by applying it to 13 additional datasets.

An in-depth description of the methodology behind nnU-Net, as well as its overarching design principles, is provided in the Methods and Supplementary Note 2, respectively. Segmentation pipelines generated by nnU-Net for all datasets are provided in Supplementary Note 6.

nnU-Net handles a wide variety of target structures and image properties. We demonstrate the value of nnU-Net as an out-of-the-box segmentation tool by applying it to 11 international biomedical image segmentation challenges comprising 23 different datasets and 53 segmentation tasks^{6,14,16–24} (<https://cremi.org/>). This selection comprised a variety of organs, organ substructures, tumors, lesions and cellular structures in 2D as well as 3D images acquired by magnetic resonance imaging (MRI), computed tomography (CT), electron microscopy (EM) and fluorescence microscopy (FM). ‘Challenges’ are international competitions that aim to assess the performance of multiple algorithms in a standardized environment⁹. In all segmentation tasks, nnU-Net was trained from scratch using only the provided challenge data. Qualitatively, we observed that nnU-Net could handle a large disparity in dataset properties and diversity in target structures; that is, generated pipeline configurations were in line with what human experts consider a reasonable or sensible setting (Supplementary Note 3, sections 1 and 2). Examples for segmentation results generated by nnU-Net are presented in Fig. 1.

nnU-Net outperforms specialized pipelines in a range of diverse tasks. Fig. 3 provides an overview of the quantitative results

achieved by nnU-Net and the competing challenge teams across all 53 segmentation tasks. Despite its generic nature, nnU-Net outperforms most existing segmentation solutions, even though the latter were specifically optimized for the respective task. Overall, nnU-Net sets a new state of the art in 33 of 53 target structures and otherwise shows performances on par with or close to the top leaderboard entries.

Details in method configuration have more impact on performance than do architectural variations. To provide a more profound picture of the current practice in deep learning-based biomedical image segmentation, we analyze as examples the participating algorithms in the recent Kidney and Kidney Tumor Segmentation (KiTS) 2019 challenge hosted by the Medical Image Computing and Computer Assisted Intervention (MICCAI) society²⁵. The MICCAI society has consistently hosted at least 50% of all annual biomedical image analysis challenges⁹. With more than 100 competitors, the KiTS challenge was the largest competition at MICCAI 2019. The first observation was that AutoML approaches were noticeably absent from the leaderboard. Only one submission (rank 18 of 100) reported the selection of “a few hyperparameters via grid search” (http://results.kits-challenge.org/miccai2019/manuscripts/peekaboo_2.pdf), while manual trial-and-error optimization represented the undeniable status quo. Notably, this observation was not specific to KiTS; we are not aware of successful submissions employing AutoML in any biomedical image segmentation competition. Fig. 4a provides an overall summary of the KiTS leaderboard (<http://results.kits-challenge.org/miccai2019>), revealing further insights on the current landscape of deep learning-based segmentation method design. First, the top 15 methods were derived from the (3D) U-Net architecture from 2016 (refs. ^{15,26}), confirming its impact on the field of biomedical image segmentation. Second, contributions using the same type of network resulted in performances spread across the entire leaderboard. Third, in examining the top 15 methods, none of the commonly used architectural modifications (for example, residual connections^{27,28}, dense connections^{29,30}, attention mechanisms³¹ or dilated convolutions^{32,33}) represented a necessary condition for good performance on the KiTS task.

Fig. 4b underscores the importance of finding good method configurations. It illustrates an analysis of algorithms that all used the same architecture variant as the challenge-winning contribution, a 3D U-Net with residual connections. While one of these methods won the challenge, other contributions based on the same principle cover the entire range of evaluation scores and rankings. Key configuration parameters were selected from respective pipeline fingerprints, illustrating the co-dependent design choices that each team made during method configuration. The drastically varying configurations submitted by contestants indicate the underlying complexity of the high-dimensional optimization problem that is implicitly posed by configuring a deep learning method for biomedical image segmentation.

nnU-Net experimentally underscores the relative importance of method configuration over architectural variations in the KiTS dataset by setting a new state of the art on the open leaderboard (nnU-Net was submitted to the leaderboard after the original challenge ended and was thus not part of the original leaderboard analysis. The methods analyzed in Fig. 4 are also listed in the open leaderboard.) with a plain 3D U-Net architecture. This observation is in line with our results from 22 additional datasets (Fig. 3).

Different datasets require different pipeline configurations. We extracted the data fingerprints of 23 biomedical segmentation datasets. As displayed in Fig. 5, this documented an exceptional dataset diversity in biomedical imaging, and revealed the fundamental reason behind the lack of out-of-the-box segmentation algorithms: the complexity of method configuration is amplified by the fact that

将这一知识提炼为成功的方法设计，可得出以下启发式规则：“将图像块大小初始化为中位数图像形状，并在适应网络拓扑（包括网络深度、沿每个轴的池化操作的数量和位置、特征图大小和卷积核大小）的同时迭代缩小它，直到在给定的GPU内存限制下网络能够以至少为二的批量大小进行训练。”所有启发式规则的详细描述见在线方法，用于推导这些规则的指导原则汇编见补充说明2。第三，我们仅将剩余的设计选择，即模型选择和后处理，设为在应用期间根据训练数据经验性地决定。我们对这一方案的实现，我们称之为nnU-Net，完全是在源自医学十项全能分割挑战赛的十个开发数据集上开发的¹⁴。

nnU-Net应用。将nnU-Net应用于新数据集时，nnU-Net的自动配置无需人工干预即可运行。因此，除了少数仍需做出的经验性选择外，不需要超出标准网络训练过程的额外计算成本。nnU-Net的自动化方法配置首先提取数据集指纹，随后执行启发式规则。默认情况下，nnU-Net生成三种不同的U-Net¹⁵配置：**二维（2D）U-Net、在全图像分辨率下运行的3D U-Net，以及3D U-Net级联**——其中第一个U-Net在下采样图像上运行，第二个U-Net被训练用于在全分辨率下细化前者生成的分割图。交叉验证后，nnU-Net根据经验选择表现最佳的配置或集成。最后，如果测量到性能提升，nnU-Net根据经验选择“非最大连通域抑制”作为后处理步骤。nnU-Net自动化配置和训练过程的输出是训练好的模型，可部署用于对未见过的图像进行预测。我们通过将nnU-Net应用于额外13个数据集，展示了编码在其固定的、基于规则的和经验的参数中的设计选择的泛化能力。

关于nnU-Net背后方法的深入描述以及其总体设计原则，分别见方法部分和补充说明2。nnU-Net为所有数据集生成的分割流水线见补充说明6。

nnU-Net能够处理多种多样的目标结构和图像属性。我们将nnU-Net应用于11个国际生物医学图像分割挑战赛来展示其作为开箱即用的分割工具的价值，这些挑战赛包含23个不同的数据集和53个分割任务^{6, 14, 16–24} (<https://cremi.org/>)。该选择涵盖了多种器官、器官子结构、肿瘤、病变和细胞结构，涉及通过磁共振成像（MRI）、计算机断层扫描（CT）、电子显微镜（EM）和荧光显微镜（FM）获取的2D及3D图像。“挑战赛”是旨在标准化环境中评估多种算法性能的国际竞赛⁹。在所有分割任务中，nnU-Net仅使用所提供的挑战赛数据从头开始训练。在定性上，我们观察到nnU-Net能够处理数据集属性上的巨大差异以及目标结构的多样性；也就是说，生成的流水线配置与人类专家认为合理或明智的设置相一致（补充说明3，第1节和第2节）。nnU-Net生成的分割结果示例见图1。

nnU-Net在一系列多样化任务中优于专门的流水线。图3提供了定量结果的概览

展示了nnU-Net与参赛竞争团队在所有53个分割任务上取得的成果。尽管nnU-Net具有通用性，它仍优于大多数现有的分割方案，即使后者是专门为各自任务优化的。总体而言，nnU-Net在53个目标结构中的33个上设定了新的最先进水平，在其他任务上则表现出与排行榜顶尖条目相当或接近的性能。

方法配置中的细节对性能的影响比架构变体更大。为了更深入地了解当前基于深度学习的生物医学图像分割实践，我们以近期由医学图像计算与计算机辅助干预（MICCAI）学会主办的肾脏与肾脏肿瘤分割（KiTS）2019挑战赛²⁵中的参赛算法为例进行分析²⁵。MICCAI学会始终主办了所有年度生物医学图像分析挑战赛中至少50%的赛事⁹。KiTS挑战赛拥有超过100名参赛者，是MICCAI 2019上规模最大的竞赛。第一个观察结果是，自动机器学习方法在排行榜上明显缺席。只有一份提交（100名中排名第18）报告了“通过网格搜索选择少数超参数”（http://results.kits-challenge.org/miccai2019/manuscripts/peekaboo_2.pdf），而手动试错优化代表了不可否认的现状。值得注意的是，这一观察并非KiTS所特有；我们不知道有任何生物医学图像分割竞赛中成功采用了自动机器学习的提交。图4a提供了KiTS排行榜的整体概况（<http://results.kits-challenge.org/miccai2019>），揭示了关于当前基于深度学习的分割方法设计格局的进一步见解。首先，排名前15的方法均源自2016年的（3D）U-Net架构（参考文献^{15,26}），证实了其对生物医学图像分割领域的影响。其次，使用相同类型网络的提交其性能分布在整个排行榜上。第三，在考察排名前15的方法时，没有任何常用的架构修改（例如，残差连接^{27,28}，密集连接^{29,30}，注意力机制³¹或空洞卷积^{32,33}）是KiTS任务上取得良好性能的必要条件。

图4b强调了寻找良好方法配置的重要性。它展示了对算法的分析，这些算法都使用了与挑战赛获胜方案相同的架构变体，即带有残差连接的3D U-Net。虽然其中一种方法赢得了挑战赛，但其他基于相同原理的方案覆盖了评估分数和排名的整个范围。关键配置参数是从各自的流水线指纹中选择的，说明了每个团队在方法配置过程中所做的相互依赖的设计选择。参赛者提交的配置差异巨大，表明为生物医学图像分割配置深度学习方法所隐含的高维优化问题具有内在复杂性。

nnU-Net通过以朴素的3D U-Net架构在公开排行榜上创造了新的最先进水平（nnU-Net是在原始挑战赛结束后才提交至排行榜的，因此未参与原始排行榜分析。图4中分析的方法也列在公开排行榜中。），从实验上强调了在KiTS数据集上方法配置相对于架构变体的相对重要性。这一观察结果与我们在另外22个数据集上的结果一致（图3）。

不同的数据集需要不同的流水线配置。我们提取了23个生物医学分割数据集的数据指纹。如图5所示，这记录了生物医学成像中异常的数据集多样性，并揭示了缺乏开箱即用的分割算法背后的根本原因：方法配置的复杂性被以下事实放大：

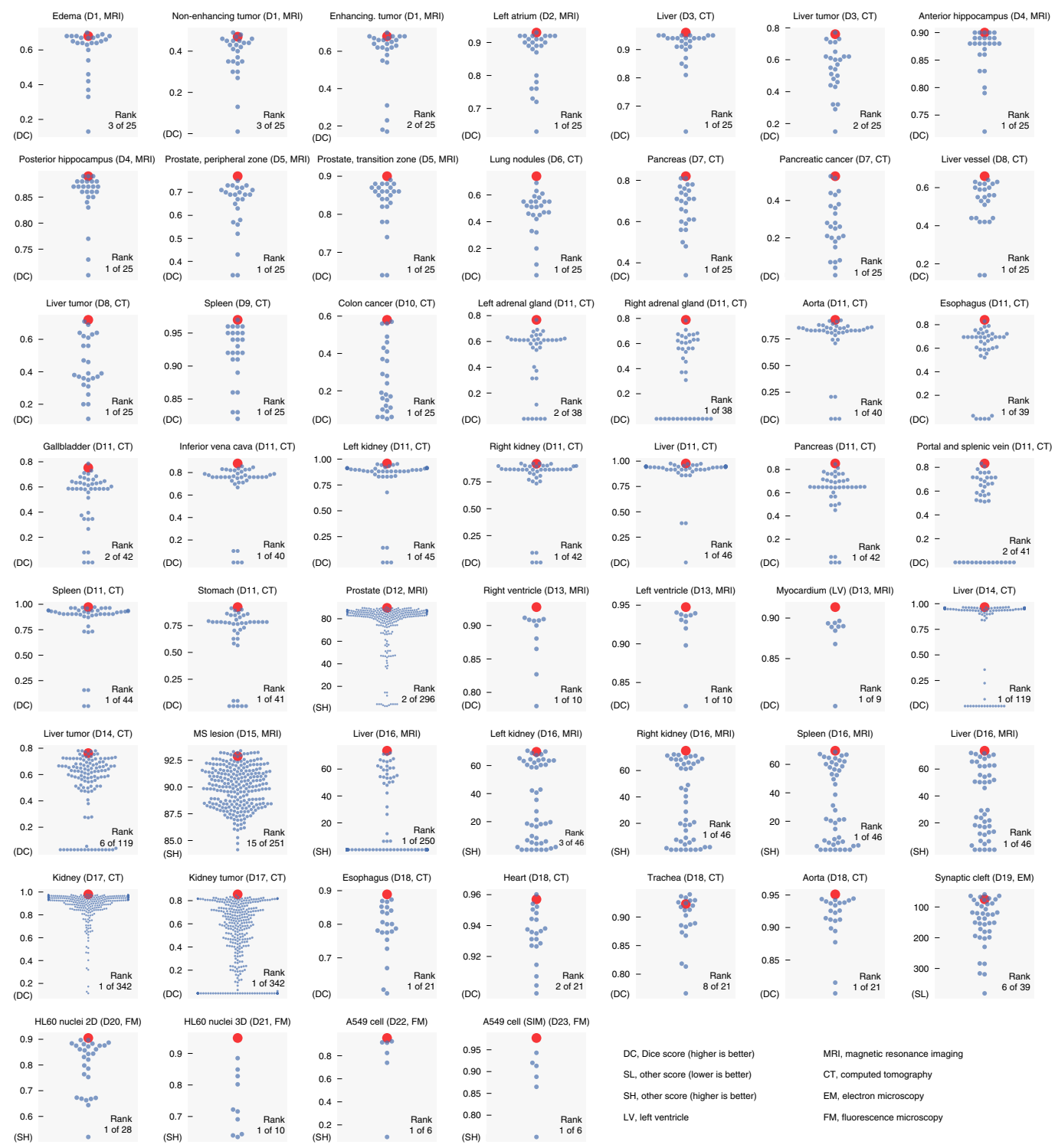


Fig. 3 | nnU-Net outperforms most specialized deep learning pipelines. Quantitative results from all international challenges that nnU-Net competed in. For each segmentation task, results achieved by nnU-Net are highlighted in red; competing teams are shown in blue. For each segmentation task, nnU-Net's rank as well as the total number of competing algorithms is displayed in the bottom-right corner of each plot. Note that for the CHAOS challenge (D16), we only participated in two of the five subtasks for reasons outlined in section 9 of Supplementary Note 6. The Cell Tracking Challenge leaderboard (D20–D23) was last accessed on 30 July 2020; all remaining leaderboards were last accessed on 12 December 2019. SIM, simulated.

suitable pipeline settings either directly or indirectly depend on the data fingerprint under potentially complex relations. As a consequence, pipeline settings that are identified as optimal for one dataset (such as KiTS, see above) may not generalize to others, resulting in a need for reoptimization for each individual dataset. nnU-Net

addresses this challenge by identifying robust design decisions and explicitly modeling key interdependencies (Fig. 2).

Multiple tasks enable robust design decisions. nnU-Net's automatic method configuration can be exploited by researchers for the

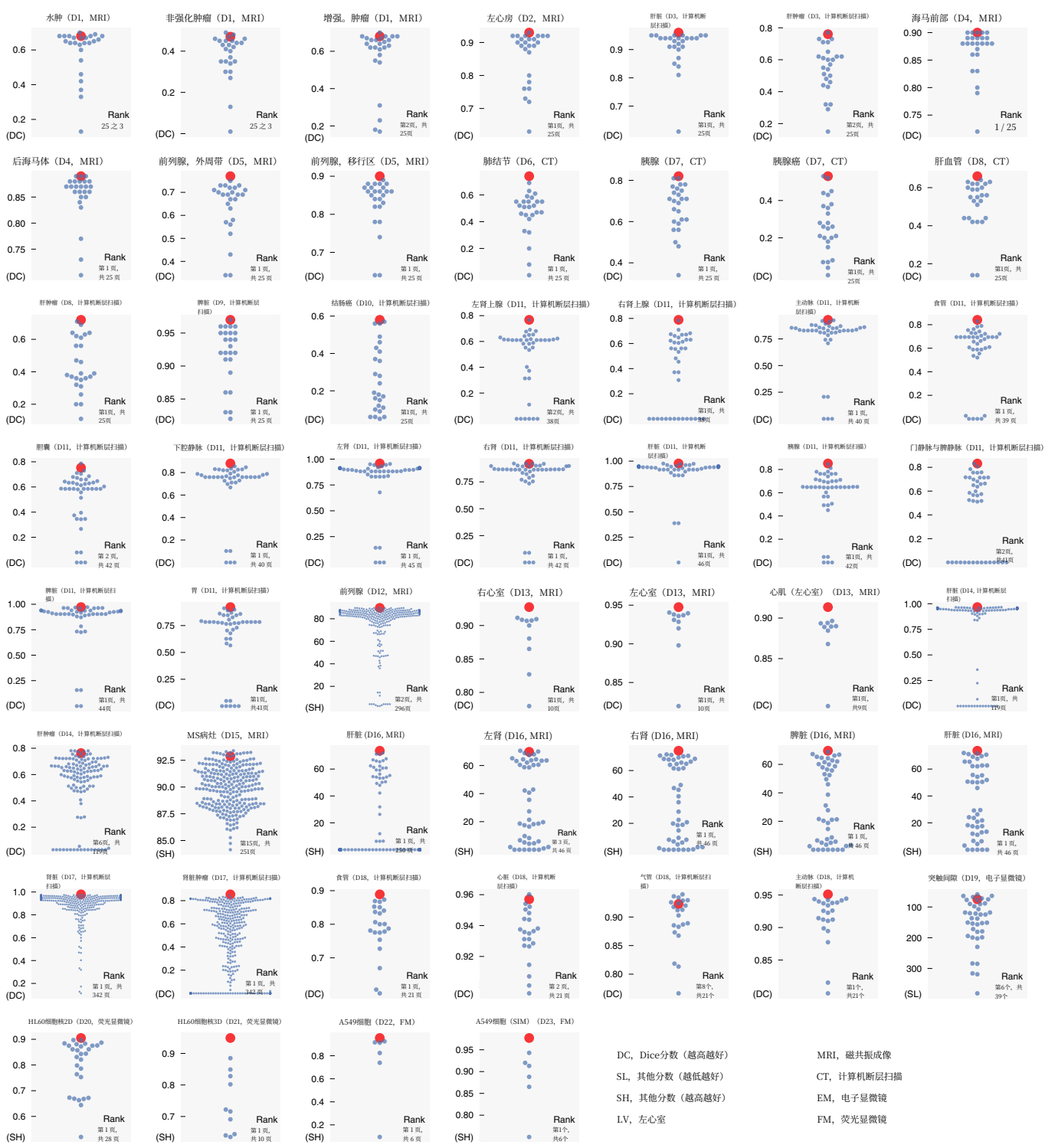


图 3 | nnU-Net 优于大多数专用的深度学习流水线。 nnU-Net 参加的所有国际挑战赛的定量结果。对于每个分割任务，nnU-Net 取得的结果以红色突出显示；竞争团队以蓝色显示。对于每个分割任务，nnU-Net 的秩以及竞争算法的总数显示在每个图的右下角。请注意，对于 CHAOS 挑战赛（D16），由于补充说明 6 第 9 节所述的原因，我们仅参加了五个子任务中的两个。细胞追踪挑战赛排行榜（D20–D23）的最后访问时间为 2020 年 7 月 30 日；所有其余排行榜的最后访问时间为 2019 年 12 月 12 日。SIM，模拟。

合适的流水线设置直接或间接依赖于数据指纹，且关系可能十分复杂。因此，对某个数据集（如上述 KiTS）而言被认定为最优的流水线设置可能无法推广到其他数据集，导致需要对每个单独的数据集重新进行优化。nnU-Net

通过识别稳健设计决策并显式建模关键相互依赖关系（图 2），解决了这一挑战。

多任务能够实现稳健的设计决策。 nnU-Net 的自动方法配置可被研究人员用于

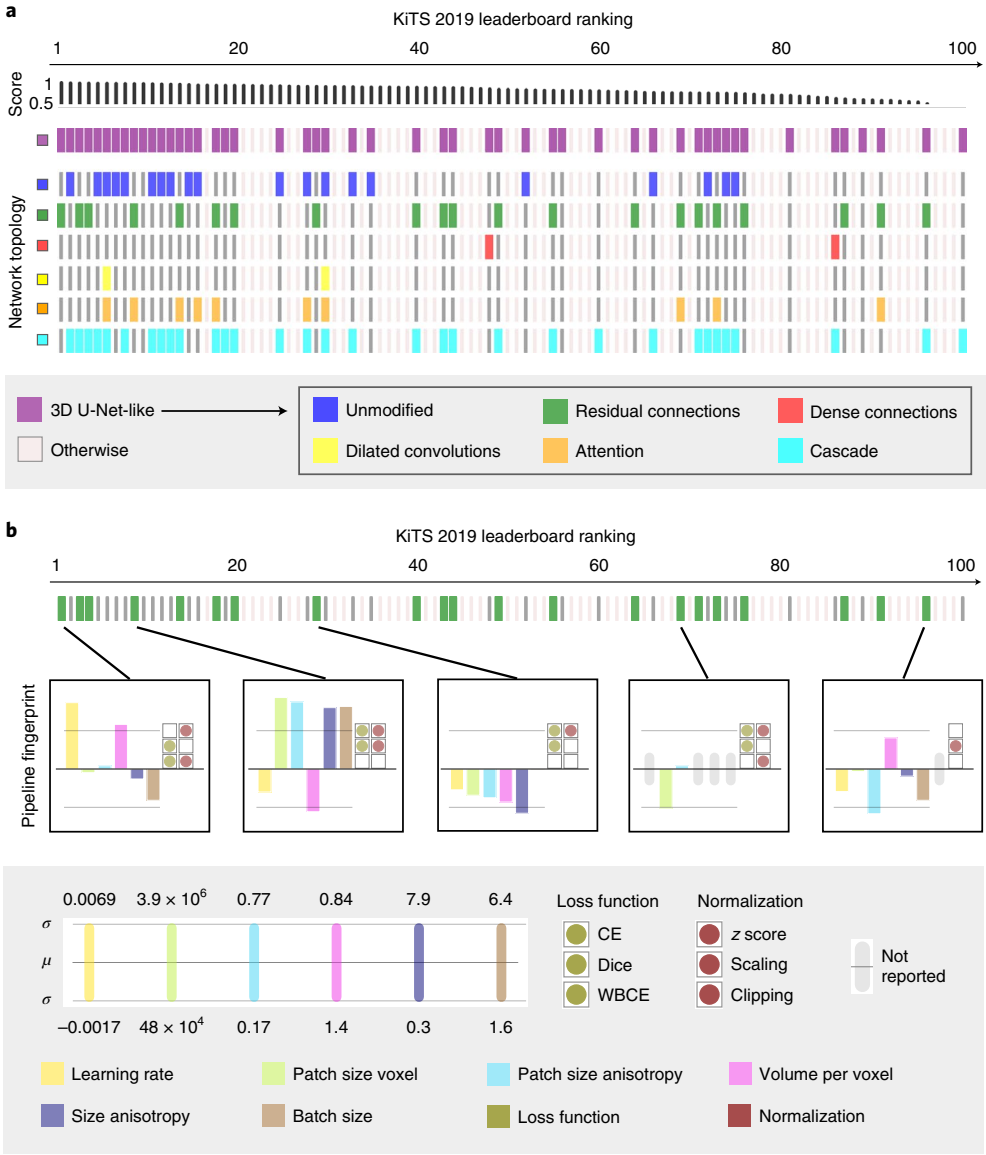


Fig. 4 | Pipeline fingerprints from KiTS 2019 leaderboard entries. **a**, Coarse categorization of leaderboard entries by architecture variation. All top 15 contributions were encoder-decoder architectures with skip-connections, 3D convolutions and output stride 1 (‘3D U-Net-like’, purple). **b**, Finer-grained key parameters selected from the pipeline fingerprints of all non-cascaded 3D U-Net-like architectures with residual connections (displayed on a z-score normalized scale). Abbreviations, CE, cross-entropy loss function; Dice, soft Dice loss function; WBCE, weighted binary CE. Further information on the KiTS 2019 challenge is available at <http://results.kits-challenge.org/>.

development of new segmentation methods. Novel ideas can easily be integrated into nnU-Net and thus tested across multiple datasets without having to manually reconfigure the entire pipeline for each dataset. To demonstrate the benefits of this approach, and also to support some of the core design choices made in nnU-Net, we systematically tested the performance of common pipeline variations by systematically modifying some of nnU-Net’s fixed parameters. The following variations were evaluated across ten different datasets and were compared against our default nnU-Net configuration, which served as a baseline in these experiments (Fig. 6).

The volatility of the ranking between datasets demonstrates how single design choices can affect segmentation performance depending on the dataset. The results clearly show that caution is required when drawing methodological conclusions from evaluations that are based on an insufficient number of datasets. While five of the nine variants achieved rank 1 in at least one of the datasets, none of them exhibited consistent improvements across the ten tasks.

The original nnU-Net configuration showed the best generalization and ranked first when results from all datasets were aggregated.

Discussion

We present nnU-Net, a deep learning-based segmentation method that automatically configures itself including preprocessing, network architecture, training and post-processing for any new task in the biomedical domain. nnU-Net sets a new state of the art for the majority of tasks on which it was evaluated, outperforming all respective specialized processing pipelines. The strong performance of nnU-Net is not achieved by a new network architecture, loss function or training scheme (hence the name nnU-Net, ‘no new net’), but by systematizing the complex process of manual method configuration, which was previously addressed either by cumbersome manual tuning or purely empirical approaches with practical limitations. We hypothesize that the reason behind nnU-Net’s state-of-the-art performance lies in the distillation of knowledge from a large pool

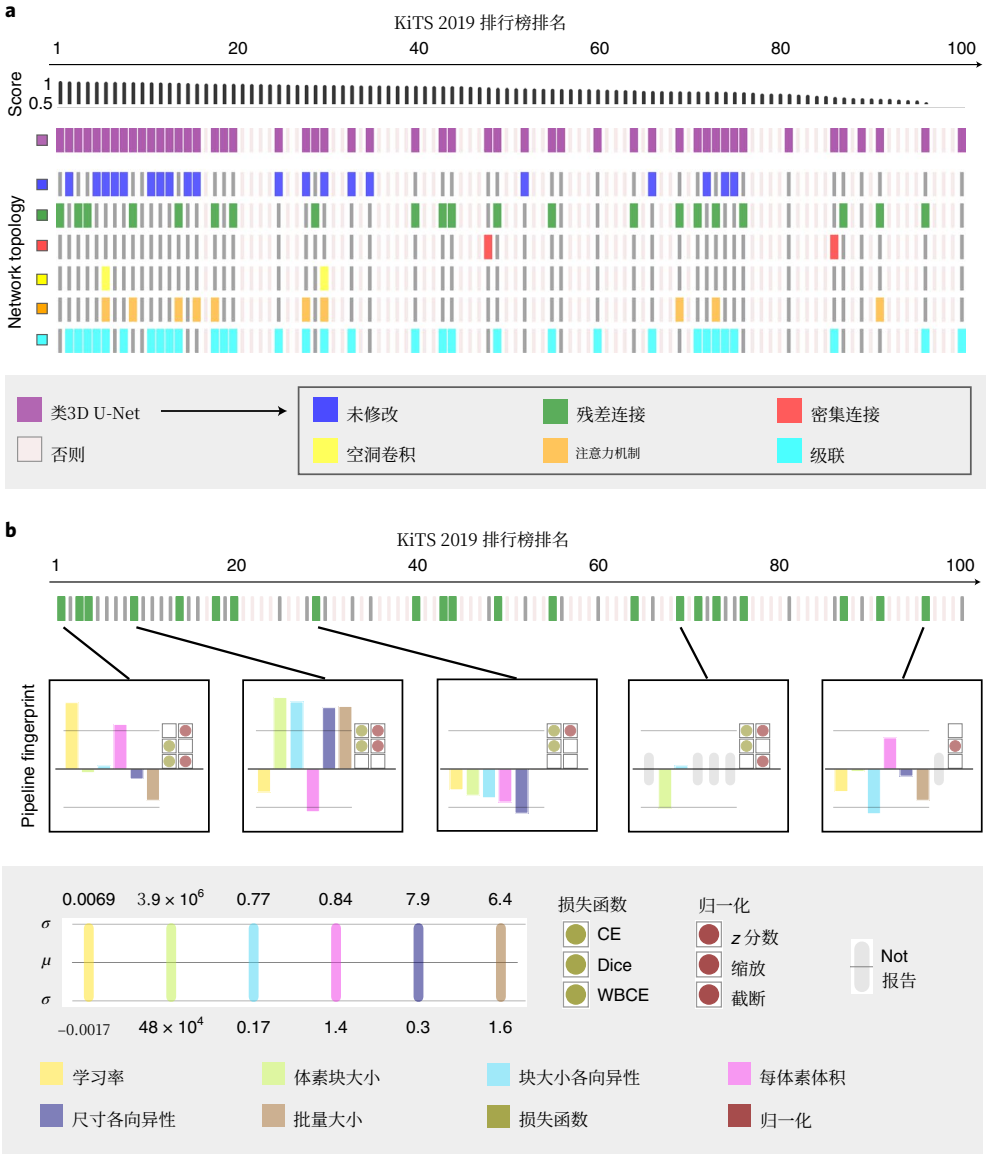


图 4 | 来自 KiTS 2019 排行榜条目的流水线指纹。 **a**, 按架构变体对排行榜条目进行的粗略分类。所有排名前 15 的提交均为具有跳跃连接、三维卷积和输出步长 1 的编码器-解码器架构（“类3D U-Net”，紫色）。 **b**, 从所有具有残差连接的非级联类3D U-Net架构的流水线指纹中选出的更细粒度关键参数（以 z 分数归一化尺度显示）。缩写：CE，交叉熵损失函数；Dice，软 Dice 损失函数；WBCE，加权二元交叉熵。有关 KiTS 2019 挑战赛的更多信息，请访问 <http://results.kits-challenge.org/>。

开发新的分割方法。新颖的想法可以轻松集成到 nnU-Net 中，从而在多个数据集上进行测试，而无需为每个数据集手动重新配置整个流水线。为了证明这种方法的优点，同时也为了支持 nnU-Net 中所做的一些核心设计选择，我们通过系统性地修改 nnU-Net 的一些固定参数，系统性地测试了常见流水线变体的性能。以下变体在十个不同的数据集上进行了评估，并与我们的默认 nnU-Net 配置进行了比较，该配置在这些实验中作为基线（图 6）。

数据集之间排名的波动性表明，单一设计选择会因数据集的不同而影响分割性能。结果清楚地显示，当基于数量不足的数据集进行评估时，从中得出方法学结论需要谨慎。虽然九个变体中有五个在至少一个数据集上达到了第1名，但它们中没有哪一个在十个任务上都表现出持续改进。

当所有数据集的结果被聚合时，原始nnU-Net配置展现出最佳的泛化能力，并排名第一。

讨论

我们提出 nnU-Net，一种基于深度学习的分割方法，能够针对生物医学领域中的任何新任务自动完成自身配置，包括预处理、网络架构、训练和后处理。nnU-Net 在大多数所评估的任务上树立了新的最先进水平，超越了所有各自的专用处理流水线。nnU-Net 的强劲性能并非来自新的网络架构、损失函数或训练方案（因此得名 nnU-Net，即“no new net”），而是来自对手动方法配置这一复杂过程的系统化，此前该过程要么通过繁琐的手动调参解决，要么通过具有实际局限性的纯经验方法解决。我们假设，nnU-Net 之所以能达到最先进性能，原因在于将来自大型数据集池的知识提炼

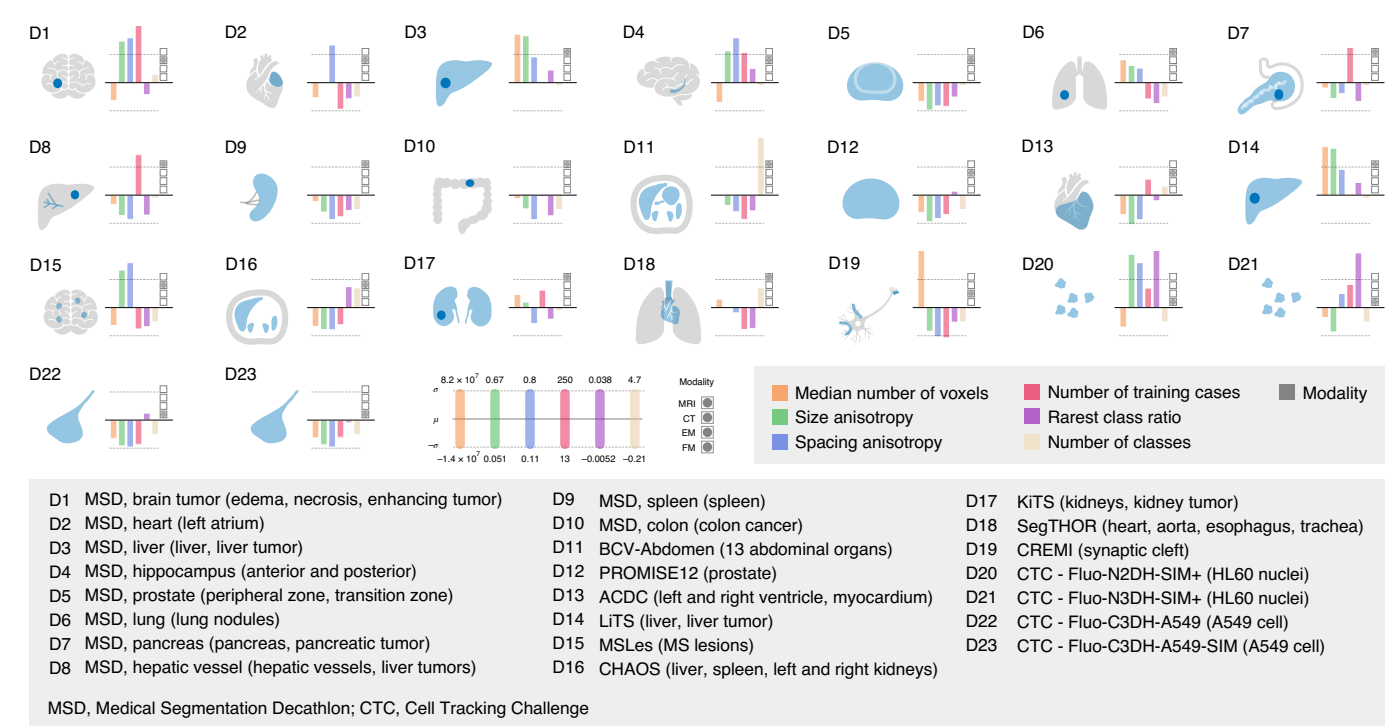


Fig. 5 | Data fingerprints across different challenge datasets. Data fingerprints show the key properties (displayed with z-score normalization over all datasets on a scale of one s.d. around the mean) for the 23 datasets used in the nnU-Net experiments (see Supplementary Note 1 for detailed dataset descriptions).

of datasets into a set of robust design choices that translates into powerful inductive biases when applied to a new dataset, and this allows for generalization abilities beyond that of models configured on a single dataset. Moreover, by condensing domain knowledge into a set of fixed, rule-based and empirical parameters, we outline a new path for automated method configuration that is computationally feasible, while at the same time covers the entire segmentation pipeline, including essential topological parameters of the network architecture. nnU-Net is a new segmentation tool that can be applied out-of-the-box, without requiring any user intervention, to a large range of biomedical imaging datasets and is thus ideal for users who require access to semantic segmentation methods and do not have the expertise, time, data or computing resources required to adapt existing solutions to their problem.

Our analysis of the KiTS leaderboard reveals a manual and not sufficiently systematic status quo of method configuration in biomedical image segmentation and highlights several implications for current research in the field. For instance, we observed that contributions using an identical type of network resulted in performances spread across the entire leaderboard (Fig. 4). This observation is in line with Litjens et al., who, in their review found that “many researchers use the exact same architectures” “but have widely varying results” (ref. ¹⁰). There are several possible reasons for why performance improvements based on architectural extensions proposed by the literature may not translate to all datasets in the domain. First, the diversity of datasets in the biomedical domain requires dedicated method configurations (Fig. 5). As a consequence, the quality of method configuration on a new dataset may overshadow the effect of the evaluated architectural modifications. This explanation is in line with an observation by Litjens et al., who concluded that “the exact architecture is not the most important determinant in getting a good solution” (ref. ¹⁰) and is supported by nnU-Net’s state-of-the-art results based on a strong method configuration in combination with a simple U-Net architecture. Second, in current research practice, evaluation is rarely performed on more than two datasets, and, even then, the datasets come with largely

overlapping properties (such as both being abdominal CT scans). As demonstrated in our multi-dataset study (Fig. 6), such evaluation is unsuitable for drawing general methodological conclusions. We relate the lack of sufficiently broad evaluations to the substantial effort required when manually adapting the configuration of proposed methods as well as existing pipelines (that is, baselines) to individual datasets. Crucially, this cumbersome process can also result in suboptimally configured baselines, causing a potential bias in literature. nnU-Net is able to alleviate these bottlenecks of current research. On the one hand, nnU-Net represents a new method that does not require manual task-specific adaptation and as such can readily serve as a strong and standardized baseline on any new segmentation task. On the other hand, nnU-Net can help to increase the number of datasets used for evaluation in the field by serving as a scalable experiment framework in which researchers easily implement methodological modifications.

While nnU-Net was shown to robustly find high quality configurations on new datasets, task-specific empirical optimization may have the potential to further improve the segmentation performance. However, as elaborated in the introduction, practical limitations of empirical AutoML approaches currently hamper their application to biomedical image segmentation. Another limitation is the lack of transparency associated with data-driven optimization (‘black box algorithms’ (ref. ¹²)) as compared to nnU-Net, for which, due to the underlying use of guiding principles, each design decision can be traced back either to certain dataset properties or a limited set of empirical experiments. Looking forward, we consider our work complementary to empirical AutoML research; nnU-Net may serve as a basis for holistic automation that can be enhanced by empirical optimization of selected design decisions such as data augmentation or network architecture.

Despite its strong performance across 53 diverse tasks, there might be segmentation tasks for which nnU-Net’s automatic adaptation is suboptimal. For example, nnU-Net was developed with a focus on the Dice coefficient as performance metric. Some tasks, however, might require highly domain-specific target metrics for

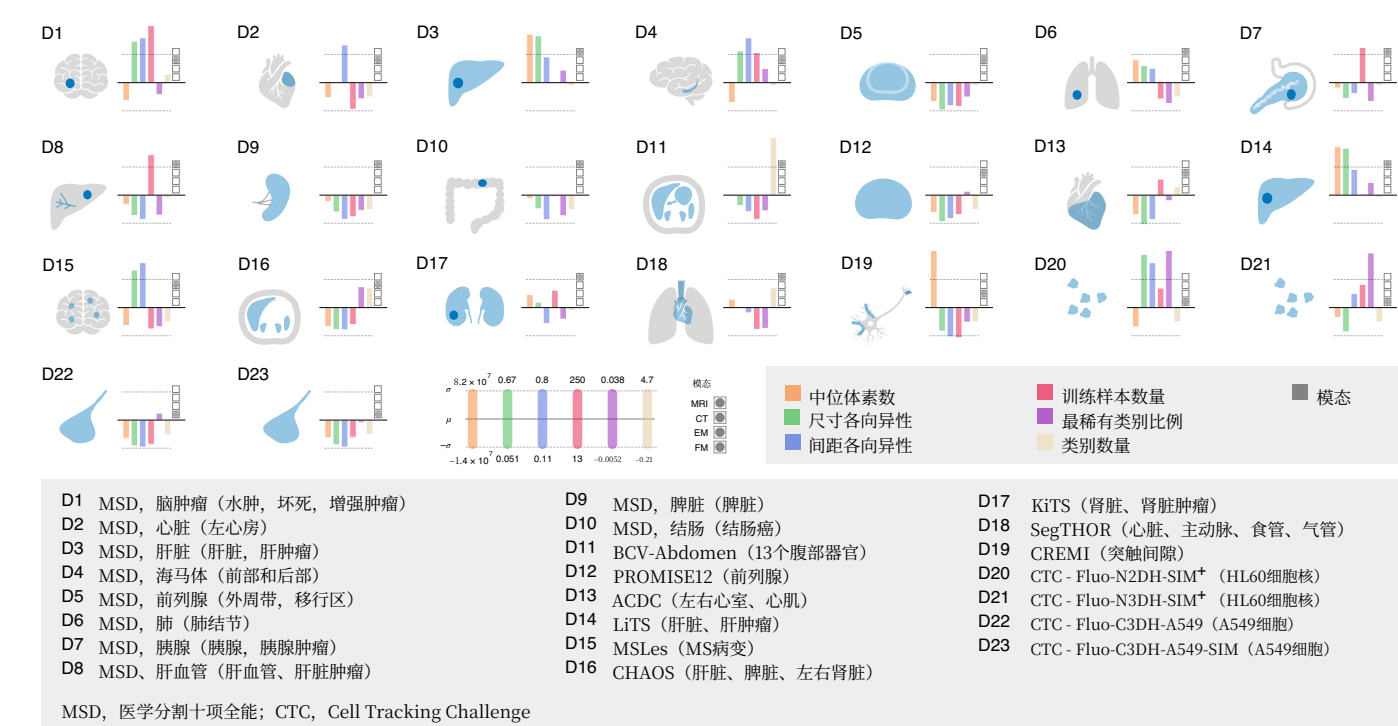


图 5 | 不同挑战数据集的数据指纹。数据指纹展示了nnU-Net实验中使用的23个数据集的关键属性（以z分数归一化显示，围绕均值的标准差范围内），数据集详细描述见补充说明1。

为一组稳健的设计选择，这些设计选择在应用于新数据集时转化为强大的归纳偏置，从而使其泛化能力超越在单一数据集上配置的模型。此外，通过将领域知识浓缩为一组固定的、基于规则的和经验性的参数，我们勾勒出一条自动化方法配置的新路径，该路径在计算上可行，同时覆盖整个分割流水线，包括网络架构中至关重要的拓扑参数。nnU-Net 是一种新的分割工具，可开箱即用，无需任何用户干预，适用于广泛的生物医学影像数据集，因此非常适合那些需要语义分割方法、却不具备将现有解决方案适配到自身问题所需的专业知识、时间、数据或计算资源的用户。

我们对KiTS排行榜的分析揭示了生物医学图像分割中方法配置的手动化且不够系统化的现状，并凸显了该领域当前研究的若干启示。例如，我们观察到使用相同类型网络的贡献在排行榜上表现参差不齐（图4）。这一观察与Litjens等人的发现一致，他们在综述中指出“许多研究人员使用完全相同的架构”“但结果差异很大”（参考文献¹⁰）。基于文献提出的架构扩展所带来的性能提升，可能无法适用于该领域所有数据集，这有多种可能的原因。首先，生物医学领域数据集的多样性需要专门的方法配置（图5）。因此，在新数据集上方法配置的质量可能掩盖所评估架构修改的效果。这一解释与Litjens等人的观察一致，他们得出结论：“确切的架构并非获得良好解决方案的最重要决定因素”（参考文献¹⁰），并且nnU-Net基于强大的方法配置结合简单的U-Net架构所取得的最先进结果也支持了这一观点。其次，在当前的研究实践中，评估很少在超过两个数据集上进行，而且即使如此，这些数据集也大多

具有重叠属性（例如均为腹部CT扫描）。正如我们的多数据集研究所证明的（图6），这种评估不适合得出普遍的方法学结论。我们将缺乏足够广泛的评估归因于手动调整所提方法以及现有流水线（即基线）的配置以适应各个数据集所需的大量工作。至关重要的是，这一繁琐的过程还可能导致基线配置次优，从而在文献中造成潜在偏差。nnU-Net能够缓解当前研究的这些瓶颈。一方面，nnU-Net代表了一种无需手动进行任务特定适配的新方法，因此可以随时作为任何新分割任务的强大且标准化的基线。另一方面，nnU-Net可以通过作为一个可扩展的实验框架来帮助增加该领域用于评估的数据集数量，研究人员可以在该框架中轻松实现方法学修改。

虽然nnU-Net已被证明能够在新数据集上稳健地找到高质量配置，但针对特定任务的经验优化可能具有进一步提升分割性能的潜力。然而，正如引言中所阐述的，经验性AutoML方法目前的实际局限性阻碍了它们在生物医学图像分割中的应用。另一个局限是与数据驱动优化（“黑箱算法”（参考文献¹²））相关的透明度不足，相比之下，nnU-Net由于在底层使用了指导原则，每个设计决策都可以追溯到特定的数据集属性或有限的实验集。展望未来，我们认为我们的工作与经验性AutoML研究是互补的；nnU-Net可以作为整体自动化的基础，并可通过针对选定设计决策（如数据增强或网络架构）的经验优化来增强。

尽管 nnU-Net 在 53 项多样化任务中表现出色，但在某些分割任务中，其自动适配可能并非最优。例如，nnU-Net 的开发侧重于以 Dice 系数作为性能指标。然而，某些任务可能需要高度领域特定的目标指标来进行评估，这可能影响方法设计。此外，可能存在尚未被考虑的数据集属性，这可能导致分割性能欠佳。一个例子是 CREMI 挑战赛中的突触间隙分割任务（

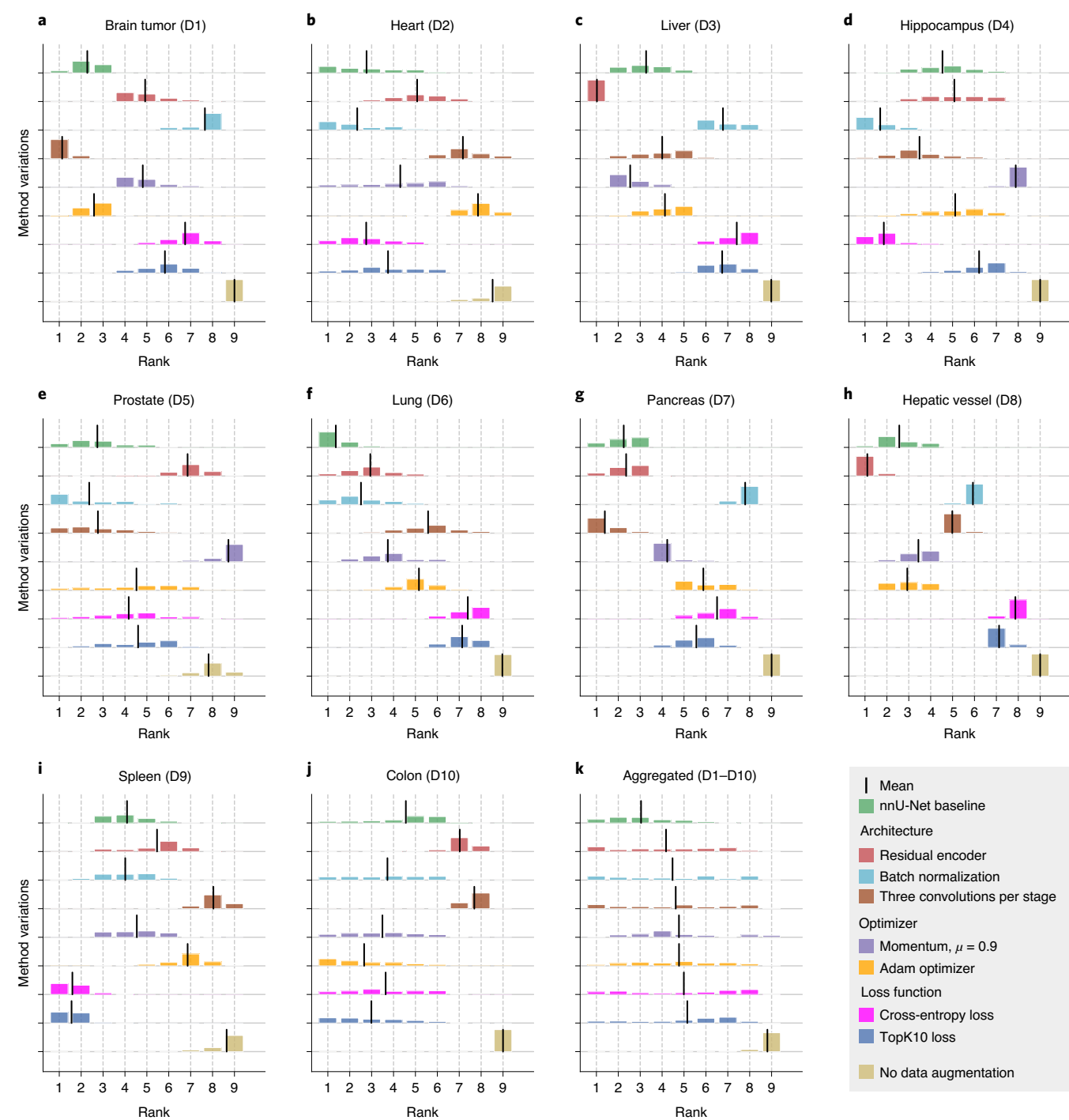


Fig. 6 | Evaluation of design decisions across multiple tasks. a–j, Evaluation of exemplary model variations on ten datasets from the medical segmentation decathlon (D1–D10, see Fig. 5 for dataset references): the application of two alternative loss functions (cross-entropy and TopK10⁴⁰), the introduction of residual connections in the encoder⁴¹, using three instead of two convolutions per resolution (resulting in a deeper network architecture), two modifications of the optimizer (a reduced momentum term and an alternative optimizer (Adam⁴²)), batch normalization⁴³ instead of instance normalization⁴⁴ and the omission of data augmentation (ablation experiments for further design choices can be found in Supplementary Note 8). Ranking stability was estimated for every dataset by aggregating validation splits of the five-fold cross-validation into one large validation set. One thousand virtual validation sets were generated via bootstrapping (drawn with replacement). Algorithms were ranked on each virtual validation set, resulting in a distribution over rankings as suggested by the challengeR tool⁴⁵. **k**, The aggregation of rankings across datasets yields insights into which design decisions robustly generalize.

evaluation, which could influence method design. Also, dataset properties that are yet unconsidered could exist, which may cause suboptimal segmentation performance. One example is the synaptic cleft segmentation task of the CREMI challenge (<https://cremi.org>). While nnU-Net’s performance is highly competitive (rank 6 of 39),

manual adaptation of the loss function, as well as EM-specific pre-processing, may be necessary to surpass state-of-the-art performance³⁴. In principle, there are two means of handling cases that are not yet sufficiently covered by nnU-Net. For potentially recurring cases, nnU-Net’s heuristics could be extended accordingly; for

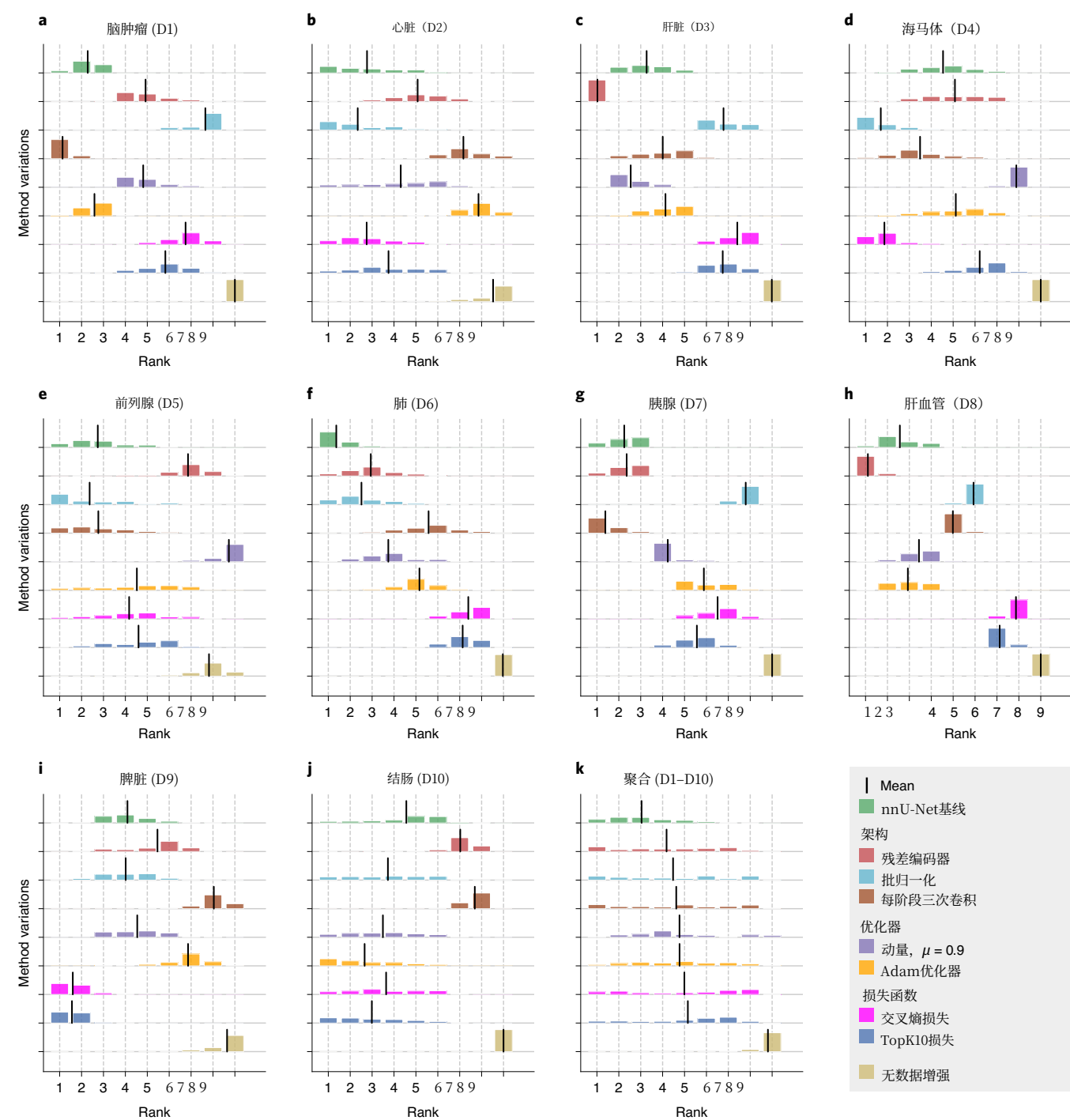


图 6 | 多任务设计决策评估。a–j，在医学分割十项全能（D1–D10，数据集参考文献见图 5）的十个数据集上评估示例性模型变体：应用两种替代损失函数（交叉熵和TopK10⁴⁰）、在编码器中引入残差连接⁴¹、在每种分辨率上使用三个而非两个卷积（导致更深的网络架构）、优化器的两种修改（减小动量项和替代优化器（Adam⁴²））、批归一化⁴³替代实例归一化⁴⁴以及省略数据增强（更多设计选择的消融实验见补充说明8）。通过将五折交叉验证的验证划分聚合为一个大型验证集，对每个数据集估计排名稳定性。通过自助法（有放回抽取）生成一千个虚拟验证集。在每个虚拟验证集上对算法进行排名，得到排名分布，如challengeR工具所建议⁴⁵。**k**，跨数据集聚合排名可揭示哪些设计决策具有稳健的泛化能力。

评估，这可能影响方法设计。此外，可能存在尚未被考虑的数据集属性，这可能导致分割性能欠佳。一个例子是 CREMI 挑战赛中的突触间隙分割任务 (<https://cremi.org>)。尽管 nnU-Net 的表现极具竞争力（在 39 个参赛者中排名第 6），

为了超越当前最先进的性能，可能需要对损失函数进行手动调整，以及针对EM的特定预处理³⁴。原则上，有两种方式可以处理nnU-Net尚未充分覆盖的情况。对于可能反复出现的情况，可以相应地扩展nnU-Net的启发式规则；对于

highly domain-specific cases, nnU-Net should be seen as a good starting point for necessary modifications.

In summary, nnU-Net sets a new state of the art in various semantic segmentation challenges and displays strong generalization characteristics requiring neither expert knowledge nor compute resources beyond standard network training. As indicated by Litjens et al. and quantitatively confirmed here, method configuration in biomedical imaging used to be considered a “highly empirical exercise”, for which “no clear recipe can be given” (ref. ¹⁰). Based on the recipe proposed in this work, nnU-Net is able to automate this often insufficiently systematic and cumbersome procedure and may thus help alleviate this burden. We propose to leverage nnU-Net as an out-of-the box tool for state-of-the-art segmentation, as a standardized and dataset-agnostic baseline for comparison and as a framework for the large-scale evaluation of novel ideas without manual effort.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41592-020-01008-z>.

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高度领域特定的情况，nnU-Net应被视为进行必要修改的良好起点。

总之，nnU-Net 在多项语义分割挑战赛中确立了新的最先进水平，并展现出强大的泛化特性，既不需要专家知识，也不需要超出标准网络训练的计算资源。正如 Litjens 等所指出的，并在本文中得到定量证实，生物医学成像中的方法配置过去被视为一种“高度经验性的练习”，对此“无法给出明确的配方”（参考文献 ¹⁰）。基于本工作提出的配方，nnU-Net 能够将这一往往缺乏系统性且繁琐的过程自动化，从而可能有助于减轻这一负担。我们建议将 nnU-Net 用作开箱即用的最先进分割工具，作为标准化且数据集无关的比较基线，以及作为无需人工投入即可大规模评估新思想的框架。

在线内容

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Methods

A quick overview of the nnU-Net design principles can be found in Supplementary Note 2. This section provides detailed information on the implementation of these principles.

Dataset fingerprints. As a first processing step, nnU-Net crops the provided training cases to their non-zero region. While this had no effect on most datasets in our experiments, it reduced the image size of brain datasets, such as D1 (brain tumor) and D15 (multiple sclerosis lesion (MSLes)), substantially and thus improved computational efficiency. Based on the cropped training data, nnU-Net creates a dataset fingerprint that captures all relevant parameters and properties: image size (that is, number of voxels per spatial dimension) before and after cropping, image spacing (that is, the physical size of the voxels), modalities (read from metadata) and number of classes for all images, as well as the total number of training cases. Furthermore, the fingerprint includes the mean and s.d., as well as the 0.5 and 99.5 percentiles of the intensity values in the foreground regions, that is, the voxels belonging to any of the class labels, computed over all training cases.

Pipeline fingerprints. nnU-Net atomates the design of deep learning methods for biomedical image segmentation by generating a so-called pipeline fingerprint that contains all relevant information. Importantly, nnU-Net reduces the design choices to the very essential ones and automatically infers these choices using a set of heuristic rules. These rules condense domain knowledge and operate on the above-described data fingerprint and project-specific hardware constraints. These rule-based parameters are complemented by fixed parameters, which are data-independent, and empirical parameters, which are optimized during training.

Fixed parameters. *Architecture template.* All U-Net architectures configured by nnU-Net originate from the same template. This template closely follows the original U-Net¹⁵ and its 3D counterpart²⁶. According to our hypothesis that a well-configured plain U-Net is still difficult to surpass, none of our U-Net configurations made use of recently proposed architectural variations, such as residual connections^{28,41}, dense connections^{29,30}, attention mechanisms³¹, squeeze-and-excitation networks⁴⁶ or dilated convolutions³². Minor changes with respect to the original architecture were made. To enable large patch sizes, the batch size of the networks in nnU-Net is small. In fact, most 3D U-Net configurations were trained with a batch size of only two (Fig. SN5.1a in Supplementary Note 5). Batch normalization⁴³, which is often used to speed up or stabilize training, does not perform well with small batch sizes^{47,48}. We therefore used instance normalization⁴⁴ for all U-Net models. Furthermore, we replaced ReLU with leaky ReLUs⁴⁹ (negative slope, 0.01). Networks are trained with deep supervision; additional auxiliary losses are added in the decoder to all but the two lowest resolutions, allowing gradients to be injected deeper into the network and facilitating the training of all layers in the network. All U-Nets employ the very common configuration of two blocks per resolution step in both the encoder and decoder, with each block consisting of a convolution, followed by instance normalization and a leaky ReLU nonlinearity. Downsampling is implemented as strided convolution (motivated by representational bottleneck⁵⁰) and upsampling is implemented as convolution transposed. As a trade-off between performance and memory consumption, the initial number of feature maps is set to 32 and doubled (halved) with each downsampling (upsampling) operation. To limit the final model size, the number of feature maps is additionally capped at 320 and 512 for 3D and 2D U-Nets, respectively.

Training schedule. Based on experience and as a trade-off between runtime and reward, all networks are trained for 1,000 epochs, with one epoch being defined as iteration over 250 mini-batches. Stochastic gradient descent with Nesterov momentum ($\mu=0.99$) and an initial learning rate of 0.01 is used for learning network weights. The learning rate is decayed throughout the training following the ‘poly’ learning rate policy⁵¹, $(1 - \text{epoch}/\text{epoch}_{\text{max}})^{0.9}$. The loss function is the sum of cross-entropy and Dice loss⁵¹. For each deep supervision output, a corresponding downsampled ground truth segmentation mask is used for loss computation. The training objective is the sum of the losses (L) at all resolutions, $L = w_1 \times L_1 + w_2 \times L_2 + \dots$. Hereby, the weights (w) are halved with each decrease in resolution, resulting in $w_2 = \frac{1}{2} \times w_1$, $w_3 = \frac{1}{4} \times w_1$, etc. and are normalized to sum to 1. Samples for the mini-batches are chosen from random training cases. Oversampling is implemented to ensure robust handling of class imbalances; 66.7% of samples are from random locations within the selected training case, while 33.3% of patches are guaranteed to contain one of the foreground classes that are present in the selected training sample (randomly selected). The number of foreground patches is rounded with a forced minimum of 1 (resulting in one random and one foreground patch with a batch size of two). A variety of data augmentation techniques are applied on the fly during training: rotations, scaling, Gaussian noise, Gaussian blur, brightness, contrast, simulation of low resolution, gamma correction and mirroring. Details are provided in Supplementary Note 4.

Inference. Images are predicted with a sliding window approach, in which the window size equals the patch size used during training. Adjacent predictions overlap by half of the size of a patch. The accuracy of segmentation decreases

toward the borders of the window. To suppress stitching artifacts and reduce the influence of positions close to borders, a Gaussian importance weighting is applied, increasing the weight of the center voxels in the softmax aggregation. Test time augmentation by mirroring along all axes is applied.

Rule-based parameters. *Intensity normalization.* There are two different image intensity normalization schemes supported by nnU-Net. The default setting for all modalities, except for CT images, is z-scoring. For this option, during training and inference, each image is normalized independently by first subtracting its mean and then dividing by its s.d. If cropping results in an average size decrease of 25% or more, a mask for central non-zero voxels is created, and normalization is applied within that mask only, ignoring the surrounding zero voxels. For CT images, nnU-Net employs a different scheme, as intensity values are quantitative and reflect physical properties of the tissue. It can therefore be beneficial to retain this information by using a global normalization scheme that is applied to all images. To this end, nnU-Net uses the 0.5 and 99.5 percentiles of the foreground voxels for clipping as well as the global foreground mean and s.d. for the normalization of all images.

Resampling. In some datasets, particularly in the medical domain, voxel spacing (the physical space that the voxels represent) is heterogeneous. Convolutional neural networks operate on voxel grids and ignore this information. To cope with this heterogeneity, nnU-Net resamples all images at the same target spacing (see paragraph below) using either third-order spline, linear or nearest-neighbor interpolation. The default setting for image data is third-order spline interpolation. For anisotropic images (maximum axis spacing ÷ minimum axis spacing > 3), in-plane resampling is performed with third-order spline interpolation, whereas out-of-plane interpolation is performed with nearest-neighbor interpolation. Treating the out-of-plane axis differently in anisotropic cases suppresses resampling artifacts, as large contour changes between slices are much more common. Segmentation maps are resampled by converting them to one-hot encodings. Each channel is then interpolated with linear interpolation, and the segmentation mask is retrieved by an argmax operation. Again, anisotropic cases are interpolated using ‘nearest neighbor’ on the low-resolution axis.

Target spacing. Selected target spacing is a crucial parameter. Larger spacings result in smaller images and thus a loss of detail, whereas smaller spacings result in larger images, preventing the network from accumulating sufficient contextual information, as the patch size is limited by the given graphics processing unit (GPU) memory budget. Although this trade-off is addressed in part by the 3D U-Net cascade (see below), a sensible target spacing for low and full resolution is still required. For the 3D full-resolution U-Net, nnU-Net uses the median value of the spacings found in the training cases computed independently for each axis as default target spacing. For anisotropic datasets, this default can result in severe interpolation artifacts or in a substantial loss of information due to large variances in resolution across the training data. Therefore, the target spacing of the lowest resolution axis is selected to be the tenth percentile of the spacings found in the training cases, if both voxel and spacing anisotropy (that is, the ratio of lowest spacing axis to highest spacing axis) are greater than three. For the 2D U-Net, nnU-Net generally operates on the two axes with the highest resolution. If all three axes are isotropic, the two trailing axes are used for slice extraction. The target spacing is the median spacing of the training cases (computed independently for each axis). For slice-based processing, no resampling along the out-of-plane axis is required.

Adaptation of network topology, patch size and batch size. Finding an appropriate U-Net architecture configuration is crucial for good segmentation performance. nnU-Net prioritizes large patch sizes while remaining within a predefined GPU memory budget. Larger patch sizes allow for more contextual information to be aggregated and thus typically increase segmentation performance. These come, however, at the cost of a decreased batch size, which results in noisier gradients during backpropagation. To improve the stability of the training, we require a minimum batch size of two and choose a large momentum term for network training (Fixed parameters). Image spacing is also considered in the adaptation process: downsampling operations may be configured to operate only on specific axes and convolutional kernels in the 3D U-Nets and may be configured to operate on certain image planes only (pseudo 2D). The network topology for all U-Net configurations is chosen on the basis of the median image size after resampling as well as the target spacing at which the images were resampled. A flow chart for the adaptation process is presented in Fig. SN5.1 in Supplementary Note 5. The adaptation of the architecture template, which is described in more detail in the following text, is computationally inexpensive. Because the GPU memory consumption estimate is based on feature map sizes, no GPU is required to run the adaptation process.

Initialization. The patch size is initialized as the median image shape after resampling. If the patch size is not divisible by 2^{n_d} ror each axis, where n_d is the number of downsampling operations, it is padded accordingly.

Architecture topology. The architecture is configured by determining the number of downsampling operations along each axis, depending on the patch size and voxel

方法

nnU-Net 设计原则的快速概览可在补充说明2中找到。本节提供了有关这些原则实施的详细信息。

数据集指纹。作为第一个处理步骤，nnU-Net 将提供的训练样本裁剪至其非零区域。虽然这在我们的实验中对大多数数据集没有影响，但它大幅减小了脑数据集（如 D1（脑肿瘤）和 D15（多发性硬化病灶 (MSLes)））的图像尺寸，从而提高了计算效率。基于裁剪后的训练数据，nnU-Net 创建一个数据集指纹，捕获所有相关参数和属性：裁剪前后的图像尺寸（即每个空间维度的体素数量）、图像间距（即体素的物理大小）、模态（从元数据读取）以及所有图像的类别数量，还有训练样本总数。此外，该指纹还包括前景区域（即属于任何类别标签的体素）中强度值的均值和标准差，以及 0.5 和 99.5 百分位数，这些均在所有训练样本上计算得出。

流水线指纹。nnU-Net通过生成所谓的流水线指纹来自动化设计用于生物医学图像分割的深度学习方法，该指纹包含所有相关信息。重要的是，nnU-Net将设计选择精简至最核心的部分，并使用一组启发式规则自动推断这些选择。这些规则浓缩了领域知识，并基于上述数据指纹和项目特定的硬件约束进行操作。这些基于规则的参数由固定参数（与数据无关）和经验参数（在训练过程中优化）补充。

固定参数。架构模板。nnU-Net配置的所有U-Net架构均源自同一模板。该模板紧密遵循原始U-Net¹⁵及其3D对应版本²⁶。根据我们的假设，即配置良好的普通U-Net仍难以超越，我们的U-Net配置均未使用近期提出的架构变体，例如残差连接^{28,41}，密集连接^{29,30}，注意力机制³¹，压缩-激励网络⁴⁶或空洞卷积³²。相对于原始架构进行了细微修改。为实现大图像块大小，nnU-Net中网络的批量大小较小。事实上，大多数3D U-Net配置仅以批量大小2进行训练（补充说明5中的图SN5.1a）。批归一化⁴³，通常用于加速或稳定训练，但在小批量大小下表现不佳^{47,48}。因此，我们对所有U-Net模型使用了实例归一化⁴⁴。此外，我们将ReLU替换为Leaky ReLU⁴⁹（负斜率，0.01）。网络采用深度监督进行训练；在解码器中，除最低的两个分辨率外，所有分辨率均添加了额外的辅助损失，使梯度能够更深入地注入网络，并促进网络中所有层的训练。所有U-Net在编码器和解码器中均采用每个分辨率步骤两个块的常见配置，每个块由卷积、实例归一化和Leaky ReLU非线性组成。下采样实现为步幅卷积（受表征瓶颈启发⁵⁰），上采样实现为转置卷积。作为性能与内存消耗之间的权衡，初始特征图数量设为32，并在每次下采样（上采样）操作时加倍（减半）。为限制最终模型大小，3D和2D U-Net的特征图数量分别额外上限为320和512。

训练计划。基于经验以及运行时间与收益之间的权衡，所有网络均训练1,000轮次，其中一轮定义为对250个小批量的迭代。采用带Nesterov动量（ $\mu=0.99$ ）的随机梯度下降进行网络权重学习，初始学习率为0.01。学习率在整个训练过程中按照‘poly’学习率策略衰减⁵¹， $(1 - \text{轮次}/\text{轮次}_{\text{max}})^{0.9}$ 。损失函数为交叉熵与Dice损失之和⁵¹。对于每个深度监督输出，使用相应的下采样真实分割掩码进行损失计算。训练目标为所有分辨率下损失（ L ）之和， $L = w_1 \times L_1 + w_2 \times L_2 + \dots$ 其中，权重（ w ）随分辨率每降低一次而减半，得到 $w_2 = 1/2 \times w_1$ ， $w_3 = 1/4 \times w_1$ ，等，并进行归一化使其总和为1。小批量的样本从随机训练样本中选取。实施过采样以确保对类别不平衡的稳健处理；66.7%的样本来自所选训练样本内的随机位置，而33.3%的图像块保证包含所选训练样本中存在的前景类别之一（随机选取）。前景图像块的数量经过取整并强制最小为1（在批量大小为二的情况下得到一个随机图像块和一个前景图像块）。训练过程中实时应用多种数据增强技术：旋转、缩放、高斯噪声、高斯模糊、亮度、对比度、低分辨率模拟、伽马校正和镜像。详细信息见补充说明4。

推理。图像通过滑动窗口方法进行预测，其中窗口大小等于训练时使用的图像块大小。相邻预测之间重叠半个图像块大小。分割精度会向窗口边界方向下降

为了抑制拼接伪影并减少靠近边界位置的影响，应用了高斯重要性加权，在softmax聚合中增加中心体素的权重。通过沿所有轴进行镜像来实现测试时增强。

基于规则的参数。强度归一化。nnU-Net 支持两种不同的图像强度归一化方案。除 CT 图像外，所有模态的默认设置为 z 评分。对于此选项，在训练和推理期间，每张图像均独立进行归一化，首先减去其均值，然后除以其标准差。如果裁剪导致平均尺寸减少 25% 或更多，则创建中心非零体素的掩膜，并仅在该掩膜内应用归一化，忽略周围的零体素。对于 CT 图像，nnU-Net 采用不同的方案，因为强度值是定量的，反映了组织的物理特性。因此，使用应用于所有图像的全局归一化方案来保留此信息可能是有益的。为此，nnU-Net 使用前景体素的 0.5 和 99.5 百分位数进行截断，并使用全局前景均值和标准差对所有图像进行归一化。

重采样。在某些数据集中，尤其是在医学领域，体素间距（体素所代表的物理空间）具有异质性。卷积神经网络在体素网格上运行，并忽略这一信息。为了应对这种异质性，nnU-Net使用三阶样条、线性或最近邻插值将所有图像重采样至相同的目标间距（见下文段落）。图像数据的默认设置是三阶样条插值。对于各向异性图像（最大轴间距 ÷ 最小轴间距 > 3），平面内重采样采用三阶样条插值，而平面外插值采用最近邻插值。在各向异性情况下对平面外轴进行不同处理可抑制重采样伪影，因为切片之间较大的轮廓变化更为常见。分割图通过将其转换为独热编码来进行重采样。然后对每个通道进行线性插值，并通过argmax操作获取分割掩码。同样，各向异性情况在低分辨率轴上使用‘最近邻’进行插值。

目标间距。选择的目标间距是一个关键参数。较大的间距会导致图像较小，从而丢失细节，而较小的间距则会导致图像较大，由于图像块大小受限于给定的图形处理单元（GPU）内存预算，网络无法积累足够的上下文信息。尽管3D U-Net级联（见下文）部分解决了这一权衡问题，但仍然需要一个合理的低分辨率和全分辨率目标间距。对于3D全分辨率U-Net，nnU-Net使用训练样本中发现的间距中值（针对每个轴独立计算）作为默认目标间距。对于各向异性数据集，由于训练数据中分辨率的巨大差异，这一默认值可能导致严重的插值伪影或大量信息丢失。因此，如果体素和间距各向异性（即最低间距轴与最高间距轴的比值）均大于三，则最低分辨率轴的目标间距选择为训练样本中发现的间距的第十百分位数。对于2D U-Net，nnU-Net通常在分辨率最高的两个轴上操作。如果三个轴均为各向同性，则使用后两个轴进行切片提取。目标间距是训练样本的间距中位数（针对每个轴独立计算）。对于基于切片的处理，无需沿平面外轴进行重采样。

网络拓扑、图像块大小与批量大小的适配。找到合适的U-Net架构配置对于良好的分割性能至关重要。nnU-Net优先选择大的图像块大小，同时保持在预定义的GPU内存预算之内。较大的图像块大小允许聚合更多的上下文信息，因此通常能提高分割性能。然而，这会以减小批量大小为代价，从而在反向传播过程中产生噪声梯度。为了提高训练的稳定性，我们要求最小批量大小为二，并为网络训练选择较大的动量项（固定参数）。图像间距在适配过程中也被考虑：下采样操作可配置为仅在特定轴上运行，3D U-Net中的卷积核可配置为仅在某些图像平面上运行（伪2D）。所有U-Net配置的网络拓扑是基于重采样后的中值图像大小以及图像被重采样时的目标间距来选择的。适配过程的流程图见补充说明5中的图SN5.1。架构模板的适配（下文将更详细地描述）在计算上是廉价的。由于GPU内存消耗估计基于特征图大小，运行适配过程不需要GPU。

初始化。图像块大小被初始化为重采样后的中位数图像形状。如果图像块大小不能被对于每个轴的 2^{n_d} 整除，其中 n_d 是下采样操作的数量，则相应地进行填充。

架构拓扑。架构的配置通过确定沿每个轴的下采样操作数量来实现，具体取决于图像块大小和体素

spacing. Downsampling is performed until further downsampling would reduce the feature map size to less than four voxels, or the feature map spacings become anisotropic. The downsampling strategy is determined by the voxel spacing; high resolution axes are downsampled separately until their resolution is within a factor of two of the lower resolution axis. Subsequently, all axes are downsampled simultaneously. Downsampling is terminated for each axis individually, once the respective feature map constraint is triggered. The default kernel size for convolutions is $3\times3\times3$ and 3×3 for 3D U-Net and 2D U-Net, respectively. If there is an initial resolution discrepancy between axes (defined as a spacing ratio larger than two), the kernel size for the out-of-plane axis is set to one until the resolutions are within a factor of two. Note that the convolutional kernel size then remains at three for all axes.

Adaptation to GPU memory budget. The largest possible patch size during configuration is limited by the amount of GPU memory. Because the patch size is initialized to the median image shape after resampling, it is initially too large to fit into the GPU for most datasets. nnU-Net estimates the memory consumption of a given architecture based on the size of the feature maps in the network, comparing it to reference values of known memory consumption. The patch size is then reduced in an iterative process, while the architecture configuration is accordingly updated in each step until the required budget is reached (Fig. SN5.1 in Supplementary Note 5). The reduction of the patch size is always applied to the largest axis relative to the median image shape of the data. The reduction in one step amounts to $2n_d$ voxels of that axis, where n_d is the number of downsampling operations.

Batch size. As a final step, the batch size is configured. If a reduction of patch size was performed, the batch size is set to two. Otherwise, the remaining GPU memory headroom is used to increase the batch size until the GPU is fully used. To prevent overfitting, the batch size is capped, such that the total number of voxels in the mini-batch do not exceed 5% of the total number of voxels of all training cases. Examples for generated U-Net architectures are presented in sections 1 and 2 of Supplementary Note 3.

Configuration of the 3D U-Net cascade. Running a segmentation model on downsampled data increases the size of patches in relation to the image and thus enables the network to accumulate more contextual information. This comes at the cost of reduced detail in the generated segmentations and may also cause errors if the segmentation target is very small or characterized by its texture. In a hypothetical scenario with unlimited GPU memory, it is thus generally favored to train models at full resolution with a patch size that covers the entire image. The 3D U-Net cascade approximates this approach by first running a 3D U-Net on downsampled images and then training a second, full-resolution 3D U-Net to refine the segmentation maps of the former. In this way, the ‘global’, low-resolution network uses maximal contextual information to generate its segmentation output, which then serves as an additional input channel to guide the second, ‘local’ U-Net. The cascade is triggered only for datasets in which the patch size of the 3D full-resolution U-Net covers less than 12.5% of the median image shape. If this is the case, the target spacing for the downsampled data and the architecture of the associated 3D low-resolution U-Net are configured jointly in an iterative process. The target spacing is initialized as the target spacing of the full-resolution data. For the patch size to cover a large proportion of the input image, the target spacing is increased stepwise by 1% while updating the architecture configuration accordingly in each step, until the patch size of the resulting network topology surpasses 25% of the current median image shape. If the current spacing is anisotropic (the difference between lowest and highest resolution axes is a factor of two), only the spacing of the higher resolution axes is increased. The configuration of the second 3D U-Net of the cascade is identical to that of the standalone 3D U-Net, the configuration process of which is described above (except that the upsampled segmentation maps of the first U-Net are concatenated to its input). Fig. SN5.1b in Supplementary Note 5 provides an overview of this optimization process.

Empirical parameters. *Ensembling and selection of U-Net configuration(s).* nnU-Net automatically determines which (ensemble of) configuration(s) to use for inference, based on the average foreground Dice coefficient computed via cross-validation on the training data. The selected model(s) can be either a single U-Net (2D, 3D full-resolution, 3D low-resolution or the full-resolution U-Net of the cascade) or an ensemble of any two of these configurations. Models are ensembled by averaging softmax probabilities.

Post-processing. Connected component-based post-processing is commonly used in medical image segmentation^{18,25}. Especially in organ image segmentation, it often helps to eliminate the detection of spurious false positives by removing all but the largest connected component. nnU-Net follows this assumption and automatically benchmarks the effect of suppressing smaller components on the cross-validation results. First, all foreground classes are treated as one component. If suppression of all but the largest region improves the average foreground Dice coefficient and does not reduce the Dice coefficient for any of the classes, this procedure is selected

as the first post-processing step. Finally, nnU-Net builds on the outcome of this step and decides whether the same procedure should be performed for individual classes.

Implementation details. nnU-Net is implemented in Python 3.8.5 using PyTorch framework 1.6.0 (ref. ³²). Batchgenerators 0.21 (ref. ³³) is used for data augmentation. Python libraries that were also used include tqdm 4.48.2, dicom2nifti 2.2.10, scikit-image 0.17.2, MedPy 0.4.0, SciPy 1.5.2, batchgenerators 0.21, NumPy 1.19.1, scikit-learn 0.23.2, SimpleITK 1.2.4 and pandas 1.1.1. For use as a framework, nnU-Net’s source code is available on GitHub (<https://github.com/MIC-DKFZ/nnUNet>). Users who wish to use nnU-Net as a standardized benchmark or to run inference with our pretrained models can install nnU-Net via PyPI. For a full description of how to use nnU-Net, as well as the most recent versions and dependencies, please refer to the online documentation available on the GitHub page.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

All 23 datasets used in this study are publicly available and can be accessed via their respective challenge websites as follows. D1–D10 Medical Segmentation Decathlon, <http://medicaldecathlon.com/>; D11 Beyond the Cranial Vault (BCV)-Abdomen, <https://www.synapse.org/#!Synapse:syn3193805/wiki/>; D12 PROMISE12, <https://promise12.grand-challenge.org/>; D13 ACDC, <https://acdc.creatis.insa-lyon.fr/>; D14 LiTS, <https://competitions.codalab.org/competitions/17094>; D15 MSLes, <https://smart-stats-tools.org/lesion-challenge>; D16 CHAOS, <https://chaos.grand-challenge.org/>; D17 KiTS, <https://kits19.grand-challenge.org/>; D18 SegTHOR, <https://competitions.codalab.org/competitions/21145>; D19 CREMI, <https://cremi.org/>; D20–D23 Cell Tracking Challenge, <http://celltrackingchallenge.net/>.

Code availability

The nnU-Net repository is available as Supplementary Software. Updated versions can be found at <https://github.com/mic-dkfz/nnunet>. Pretrained models for all datasets used in this study are available for download at <https://zenodo.org/record/3734294>.

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间距。下采样会持续进行，直到进一步下采样会将特征图大小缩减至小于四个体素，或特征图间距变为各向异性。下采样策略由体素间距决定；高分辨率轴会单独进行下采样，直到其分辨率与低分辨率轴的差距在两倍以内。随后，所有轴同时进行下采样。一旦触发各自的特征图约束，每个轴的下采样便单独终止。3D U-Net 和 2D U-Net 的卷积默认卷积核大小分别为 $3\times3\times3$ 和 3×3 。如果各轴之间存在初始分辨率差异（定义为间距比大于二），则平面外轴的卷积核大小设置为 1，直到分辨率差距在两倍以内。需要注意的是，此时所有轴的卷积核大小均保持为 3。

适应**GPU**内存预算。 配置期间可能的最大图像块大小受GPU内存量的限制。由于图像块大小在重采样后被初始化为中位数图像形状，对于大多数数据集而言，它最初过大而无法装入GPU。nnU-Net根据网络中特征图的大小估算给定架构的内存消耗，并将其与已知内存消耗的参考值进行比较。然后通过迭代过程减小图像块大小，同时每一步相应地更新架构配置，直到达到所需的预算（补充说明5中的图SN5.1）。图像块大小的缩减始终应用于相对于数据中位数图像形状的最大轴。每一步的缩减量相当于该轴的 $2n_d$ 个体素，其中 n_d 是下采样操作的次数。

批量大小。 作为最后一步，配置批量大小。如果执行了图像块大小的缩减，则将批量大小设为二。否则，剩余的GPU内存余量用于增加批量大小，直到GPU被充分利用。为防止过拟合，批量大小设有上限，使得小批量中的体素总数不超过所有训练样本体素总数的5%。生成的U-Net架构示例见补充说明3的第1节和第2节。

3D U-Net级联的配置。在下采样数据上运行分割模型会增加图像块相对于图像的大小，从而使网络能够积累更多的上下文信息。但代价是生成的分割图中细节减少，如果分割目标非常小或以其纹理为特征，也可能导致错误。在GPU内存无限的假设场景中，通常倾向于在全分辨率下训练模型，图像块大小覆盖整个图像。3D U-Net级联通过首先在下采样图像上运行3D U-Net，然后训练第二个全分辨率3D U-Net来细化前者的分割图，从而近似实现这种方法。通过这种方式，“全局”低分辨率网络利用最大化的上下文信息来生成其分割输出，该输出随后作为额外的输入通道来引导第二个“局部”U-Net。仅当3D全分辨率U-Net的图像块大小覆盖中位数图像形状的不到12.5%时，才会触发级联。在这种情况下，下采样数据的目标间距和相关3D低分辨率U-Net的架构会在迭代过程中联合配置。目标间距初始化为全分辨率数据的目标间距。为使图像块大小覆盖输入图像的较大比例，目标间距以1%的步长逐步增加，同时在每一步相应地更新架构配置，直到所得网络拓扑的图像块大小超过当前中位数图像形状的25%。如果当前间距是各向异性的（最低和最高分辨率轴之间的差异为两倍），则仅增加较高分辨率轴的间距。级联中第二个3D U-Net的配置与独立3D U-Net的配置相同，其配置过程如上所述（不同之处在于第一个U-Net的上采样分割图被拼接到其输入端）。补充说明5中的图SN5.1b提供了此优化过程的概述。

经验参数。 *U-Net*配置的集成与选择。 nnU-Net基于训练数据上通过交叉验证计算的平均前景Dice系数，自动确定用于推理的配置（集成）。所选模型可以是单个U-Net（2D、3D全分辨率、3D低分辨率或级联中的全分辨率U-Net），也可以是这些配置中任意两个的集成。模型通过平均softmax概率进行集成。

后处理。基于连通分量的后处理常用于医学图像分割^{18,25}。尤其在器官图像分割中，它通常有助于通过仅保留最大连通分量来消除对虚假阳性的检测。nnU-Net遵循这一假设，并自动评估抑制较小分量对交叉验证结果的影响。首先，所有前景类别被视为一个连通分量。如果抑制除最大区域外的所有区域能够提高平均前景Dice系数，且不降低任何类别的Dice系数，则选择该流程

作为第一个后处理步骤。最后，nnU-Net基于该步骤的结果，决定是否应对各个类别执行相同的流程。

实现细节。 nnU-Net使用Python 3.8.5和PyTorch框架1.6.0实现（参考文献³²）。数据增强使用Batchgenerators 0.21（参考文献³³）。其他使用的Python库包括tqdm 4.48.2, dicom2nifti 2.2.10, scikit-image 0.17.2, MedPy 0.4.0, SciPy 1.5.2, batchgenerators 0.21, NumPy 1.19.1, scikit-learn 0.23.2, SimpleITK 1.2.4和pandas 1.1.1。作为框架使用时，nnU-Net的源代码可在GitHub上获取（<https://github.com/MIC-DKFZ/nnUNet>）。希望将nnU-Net用作标准化基准或使用我们预训练模型进行推理的用户，可以通过PyPI安装nnU-Net。有关如何使用nnU-Net的完整说明以及最新版本和依赖项，请参阅GitHub页面上的在线文档。

报告摘要。有关研究设计的更多信息，请参阅与本文相关的Nature Research Reporting Summary。

数据可用性

本研究使用的全部23个数据集均为公开可用，可通过各自对应的挑战赛网站访问，具体如下。D1–D10 Medical Segmentation Decathlon, <http://medicaldecathlon.com/>; D11 Beyond the Cranial Vault (BCV)-Abdomen, <https://www.synapse.org/#!Synapse:syn3193805/wiki/>; D12 PROMISE12, <https://promise12.grand-challenge.org/>; D13 ACDC, <https://acdc.creatis.insa-lyon.fr/>; D14 LiTS, <https://competitions.codalab.org/competitions/17094>; D15 MSLes, <https://smart-stats-tools.org/lesion-challenge>; D16 CHAOS, <https://chaos.grand-challenge.org/>; D17 KiTS, <https://kits19.grand-challenge.org/>; D18 SegTHOR, <https://competitions.codalab.org/competitions/21145>; D19 CREMI, <https://cremi.org/>; D20–D23 Cell Tracking Challenge, <http://celltrackingchallenge.net/>。

代码可用性

nnUNet代码库作为补充软件提供。更新版本可在<https://github.com/mic-dkfz/nnunet>获取。本研究中使用的所有数据集的预训练模型可在<https://zenodo.org/record/3734294>下载。

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Author contributions

F.I. and P.F.J. conceptualized the method and planned the experiments with the help of S.A.A.K., J.P. and K.H.M.-H. F.I. implemented and configured nnU-Net and conducted the experiments on the 23 selected datasets. F.I. and P.F.J. analyzed the results and performed the KiTS analysis. P.F.J., S.A.A.K. and K.H.M.-H. conceived the communication and presentation of the method. P.F.J. designed and created the figures.

P.F.J., F.I. and K.H.M.-H. wrote the paper with contributions from J.P. and S.A.A.K. K.H.M.-H. managed and coordinated the overall project. S.A.A.K. started work on this research as a PhD student at the German Cancer Research Center.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to K.H.M.-H.

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F.I. 和P.F.J.在S.A.A.K.、J.P.和K.H.M.-H.的帮助下, 构思了方法并规划了实验。F.I.实现并配置了nnU-Net, 并在23个选定数据集上进行了实验。F.I.和P.F.J.分析了结果并进行了KiTS分析。P.F.J.、S.A.A.K.和K.H.M.-H.构思了方法的沟通和展示方式。P.F.J.设计并制作了图表。

P.F.J.、F.I.和K.H.M.-H.撰写了论文, J.P.和S.A.A.K.作出了贡献。K.H.M.-H.管理和协调了整个项目。S.A.A.K.在德国癌症研究中心攻读博士学位期间开始了这项研究。

利益冲突

作者声明无利益冲突。

附加信息

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通讯及材料索取请联系K.H.M.-H.**同行评审信息** Rita Strack是本文的主要编辑, 负责管理本文的编辑流程, 并与编辑团队其他成员合作完成同行评审。**转载和许可信息**可在以下网址获取: www.nature.com/reprints。

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Data exclusions	No data was excluded from the analysis.
Replication	All reported results are based on test sets from external challenges. Thus, scientific best practice dictates to perform only one submission (i.e. one replication) to avoid overfitting. During algorithm training we have not observed failed trainings or configurations, i.e. all attempted trainings were successful. The ablation studies in Figure 6 are reported as cross-validation results (1 run) from which 1000 bootstrap subsets were sampled to ensure ranking stability.
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